Maps and Geospatial Data

Choosing spatial representations that match the question

Information Visualization · Lecture 16 · Week 9

Agenda

01	**Spatial data**	Fields, geometry, map encodings, and when a map helps.
02	**Common map layouts**	Choropleths, symbol maps, cartograms, and grid maps.
03	**MAUP**	Spatial aggregation, zone and scale effects, and gerrymandering.
04	**Appendix**	Coordinates, projection families, and what each projection preserves or distorts.

PART 01

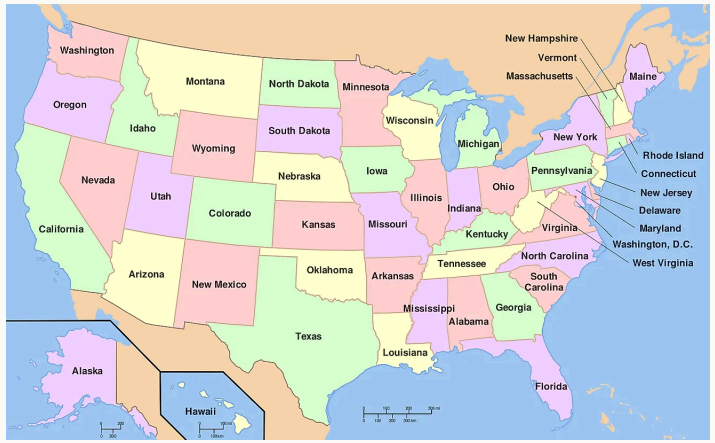
Which spatial data uses a map?

Spatial fields describe values across space. Geographic geometry describes the shapes and locations that a map displays.

Map layouts often work with spatial data

Spatial data links data records to positions or geometry in space.

State population data

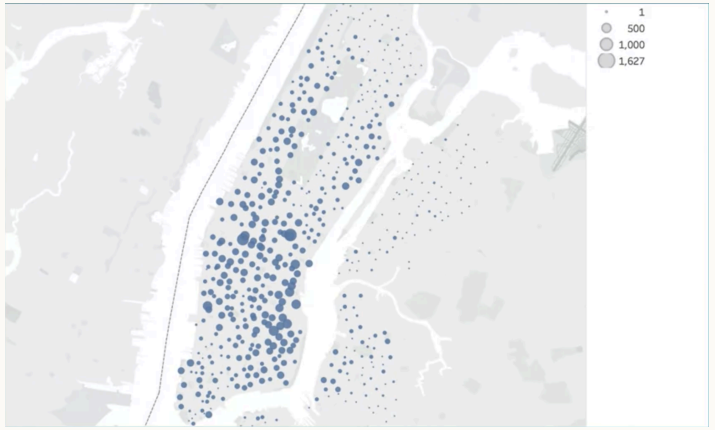


State boundary coordinates
+ state names and population.

Brain MRI data

Voxel positions
+ measured MRI signal intensities.

Citi Bike data



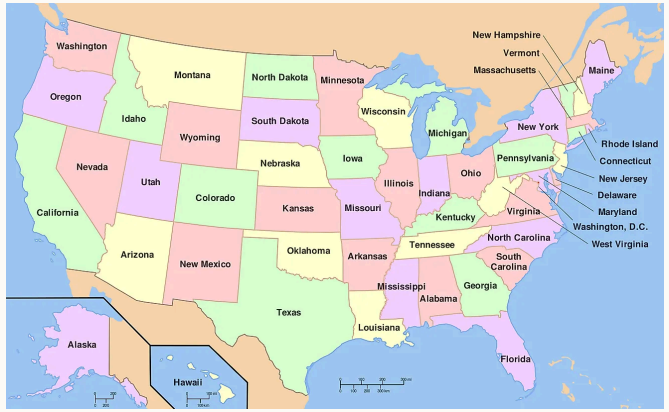
Station coordinates
+ station IDs, names, and trip records.

When does spatial data need a map?

Spatial fields

A spatial field associates each location in a domain with a value. MRI data links voxel positions to measured signal intensities.

Non-geographic fields **usually do not need a map**. A slice or volume rendering can show values within the body. Geographic fields, such as temperature across a country, can also use maps.

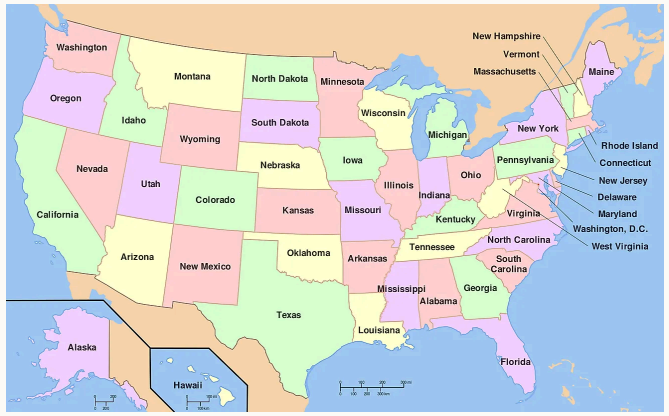


Geometry data

Object-based spatial data links each object's geometry with its attributes. A state record combines boundary coordinates, a state name, and population.

Geographic geometry **usually needs a map** when locations, shapes, or spatial relationships matter to the task.

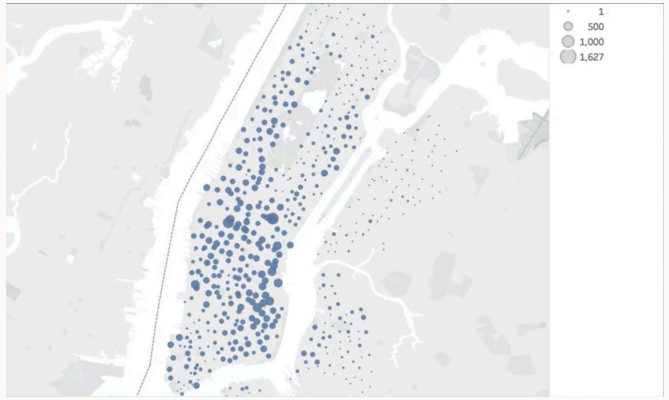
Two types of geographic geometry data



Spatial objects

Spatial objects have **shapes or boundaries that matter** to the task.

States and lakes use **area marks** that follow their boundaries. Rivers and roads use **line marks** that follow their paths.



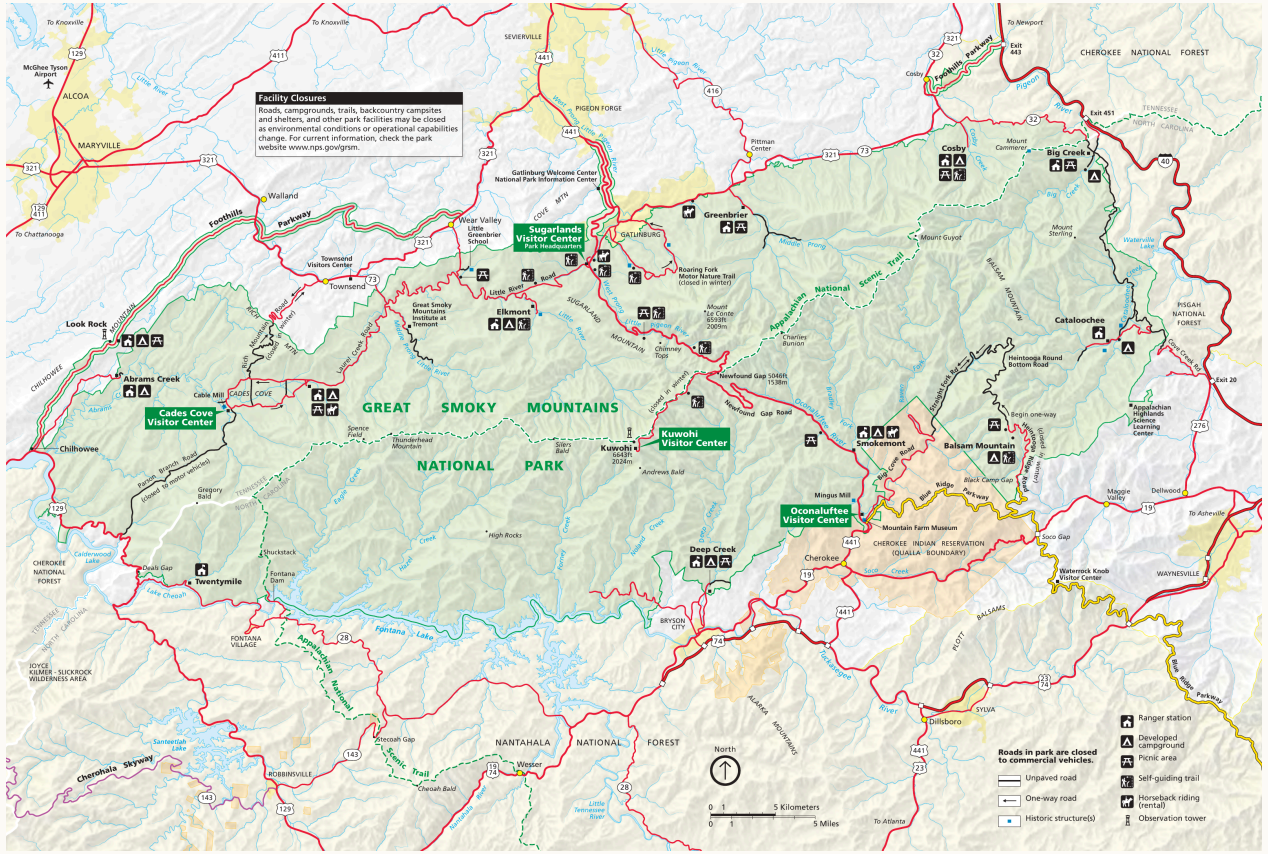
Geographically located non-spatial objects

These objects have geographic locations, but their **physical shapes are not represented**.

Citi Bike stations use **point marks** at their coordinates. The circles do not show station footprints. The map provides geographic context.

Spatial objects use marks that follow their shape

Spatial objects are represented by their geographic shape, boundary, or extent because those properties are relevant to the task.



Map: National Park Service · Great Smoky Mountains National Park.

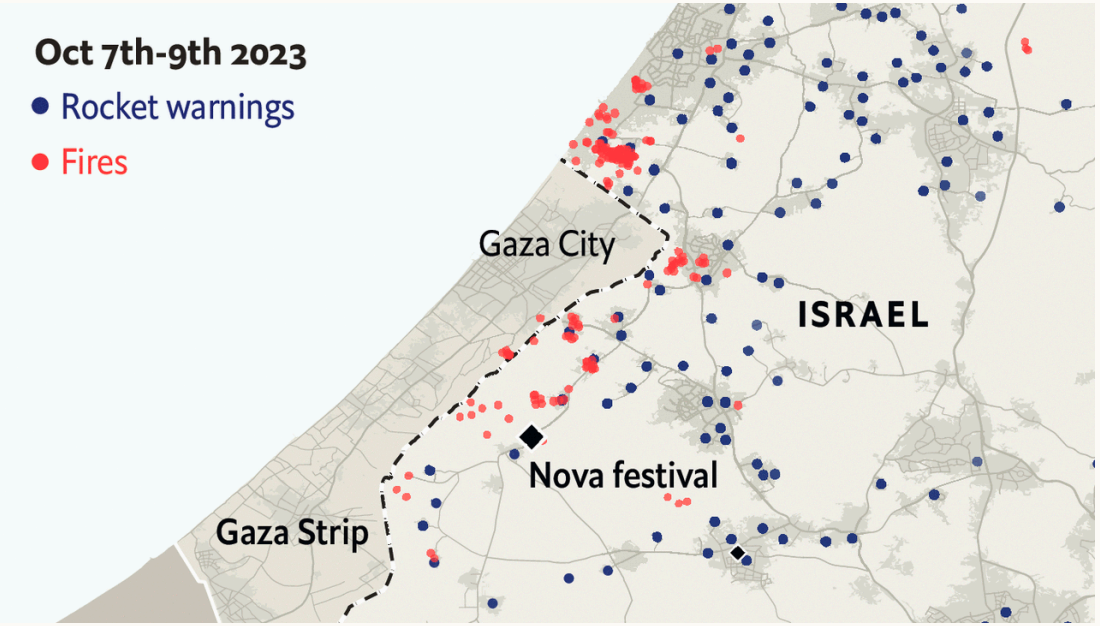
Great Smoky Mountains National Park

Area marks follow object boundaries. The green park region and blue Fontana Lake show each object's geographic shape.

Line marks follow object paths. Blue rivers and red roads trace their routes, including Newfound Gap Road.

Located objects use point marks on a map

Non-spatial objects that are geographically located are represented by a location rather than by their physical shape or extent.



In this map of rocket warnings and fires:

Point marks show where events were recorded. Blue points locate rocket warnings, and red points locate fires.

The points' shape and size **do not represent the events' shape or area of impact**.

The background map provides geographic context.

Image: BBC News · Coverage of the Hamas attack on the Nova music festival, October 2023.

Marks and channels in a map

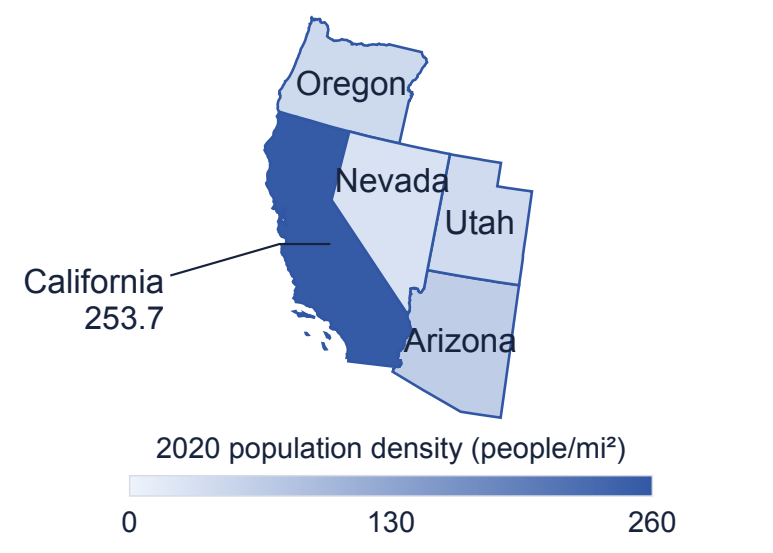
When spatial data is shown on a map, **marks encode spatial information**. Their other **visual channels encode additional attributes**.

Marks: spatial information

One area mark = one state

Each state is an area mark positioned and shaped by its boundary coordinates.

Channels: other attributes



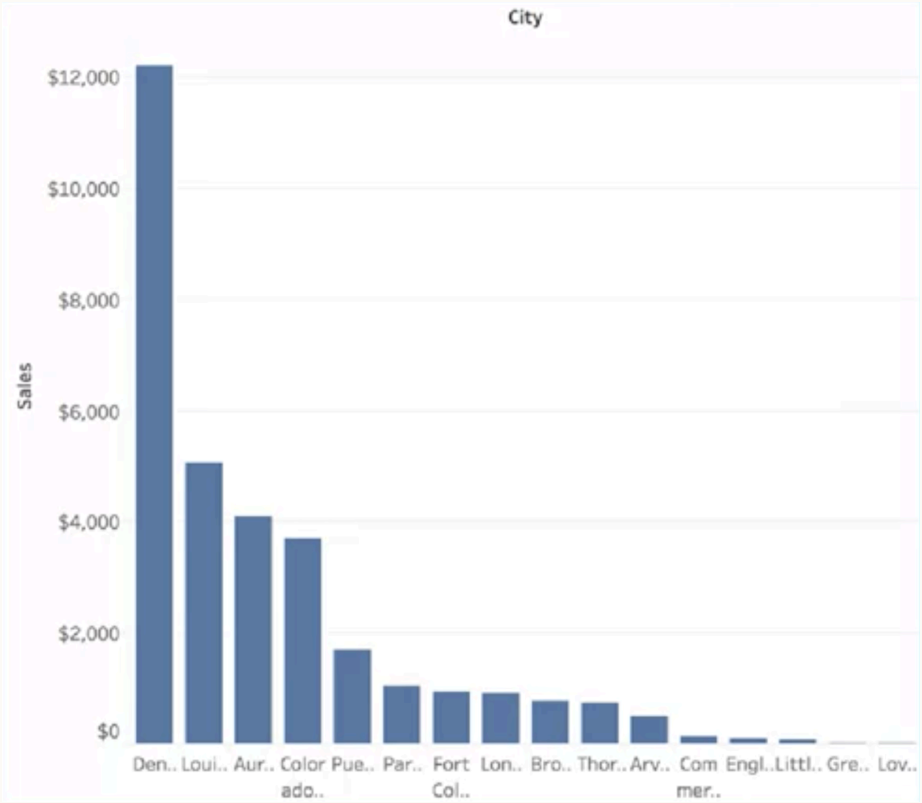
The color channel encodes population density on those same area marks.

Data: U.S. Census Bureau, 2020 population density · Boundaries: us-atlas / U.S. Census Bureau, 2017.

When do we need a map? It depends on the task.

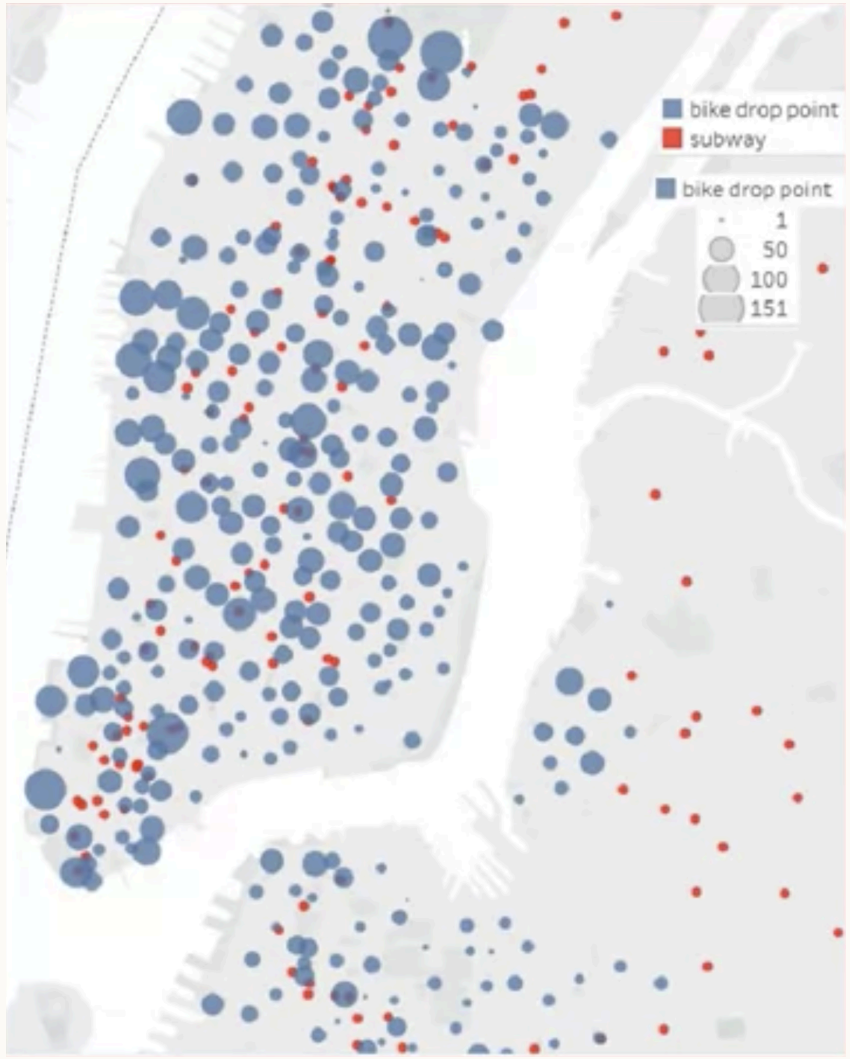
Chocolate sales in Colorado cities.

Task: Understanding how the volume of sales changes / which city has the biggest sales



Task: Exploring if there is a spatial relation between sales

Maps help answer questions about proximity



Task: Are Citi Bike drop-offs close to subway stations?

Blue circles show bike drop-off locations. Red squares show subway stations.

A map helps us explore **proximity and spatial overlap** because both sets of locations share the same geographic reference.

Geographic data does not always need a map

Graphics: Alan Smith and Laura Noonan / Financial Times · Data: FT research · 2016.

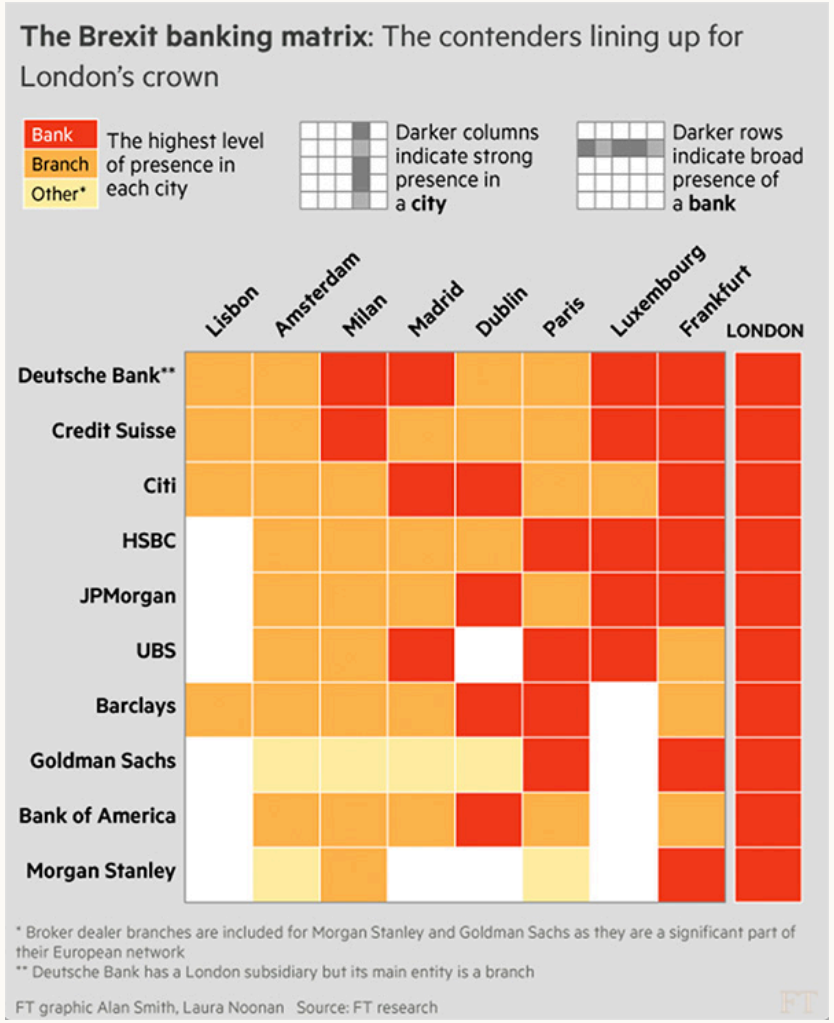
This historical Brexit example compares ten banks in London and eight other European cities.

Question: which cities have a bank-presence profile most similar to London's?

The task compares **bank presence, not locations or distances**.

The map scatters bank lists across the page. Readers must scan between cities and remember the entries to compare them.

A matrix makes bank–city comparisons easier



Graphics: Alan Smith and Laura Noonan / Financial Times · Data: FT research · 2016.

The task is to compare **financial-industry similarity**, represented here by cities' bank-presence profiles, rather than geographic proximity.

Each column is a city and each row is a bank. Unlike the map's scattered lists, **the same bank stays on the same row**, so cities can be compared directly.

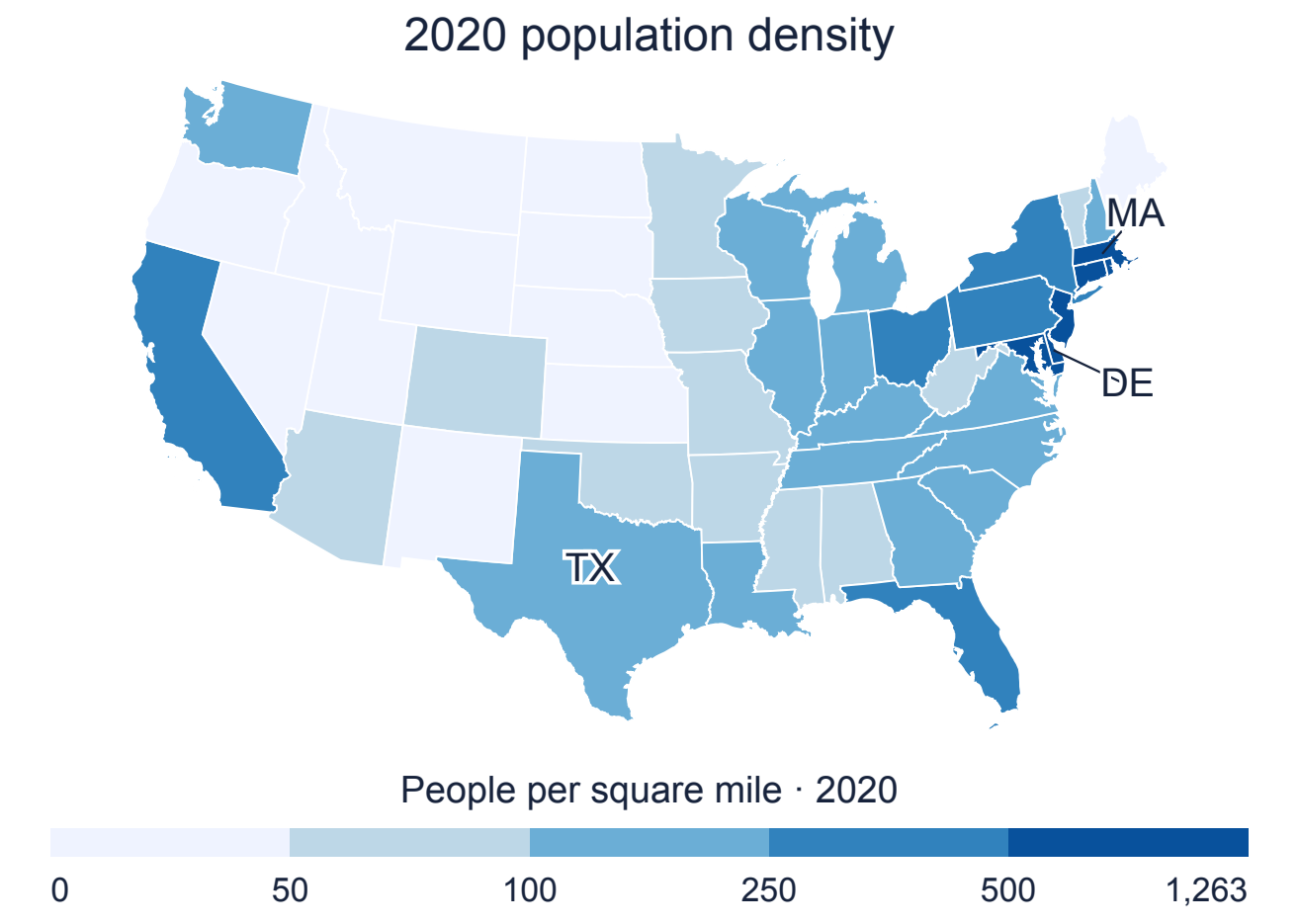
Frankfurt matches London's red “Bank” category in 7 of 10 rows. Lisbon has four orange “Branch” entries and six blank cells. Frankfurt therefore has the more similar bank-presence profile.

PART 02

Common map layouts

Choropleths, symbol maps, and cartograms encode values differently. Each makes some comparisons easier and introduces its own limitations.

A choropleth colors each region by its value



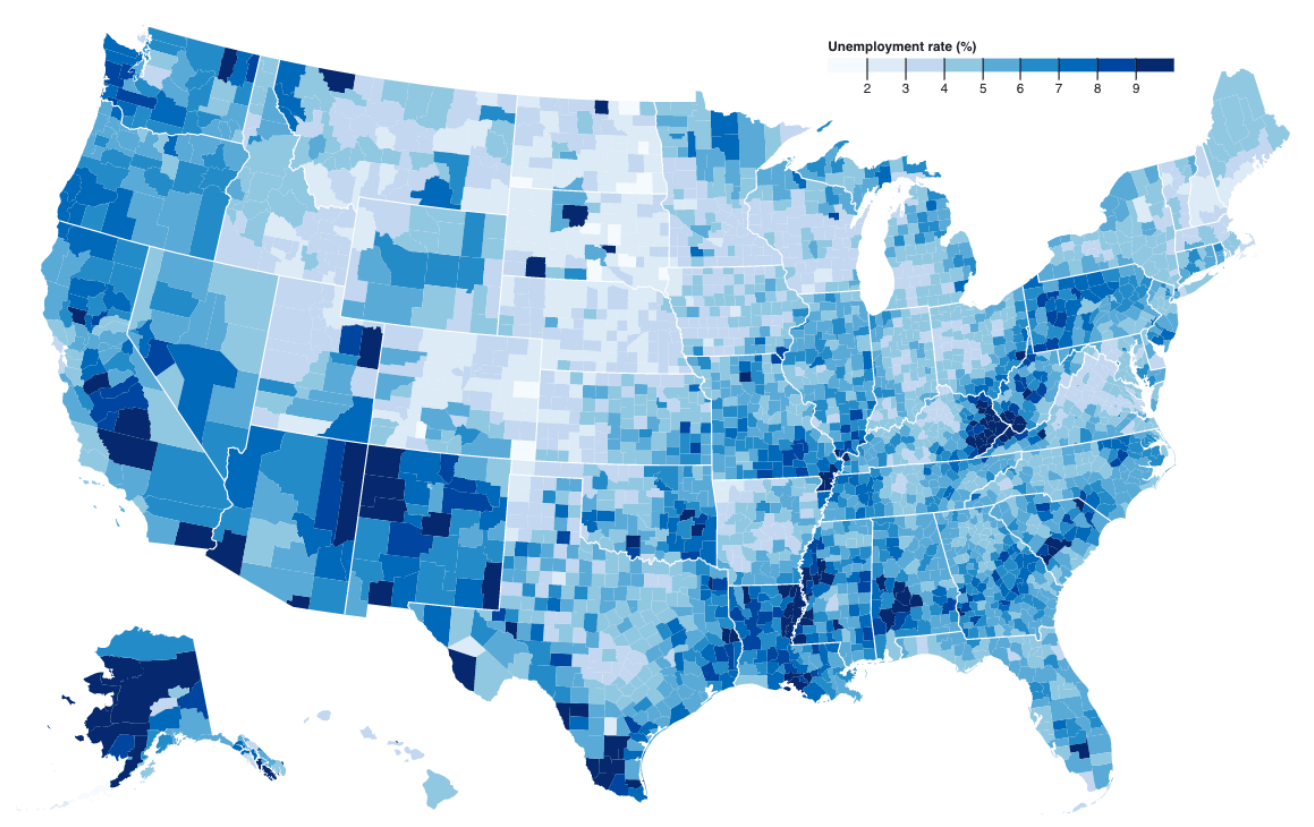
A choropleth map **uses each geographic region's color to encode its attribute value**.

Each predefined region's boundary defines an area mark. Ordered color represents the quantitative attribute.

Here, each state has one population-density value. Read the legend to convert its color into a value range.

Data: U.S. Census Bureau · 2020 · Boundaries: us-atlas / Census 2017 · Contiguous 48 states.

Choropleths make spatial patterns easy to see



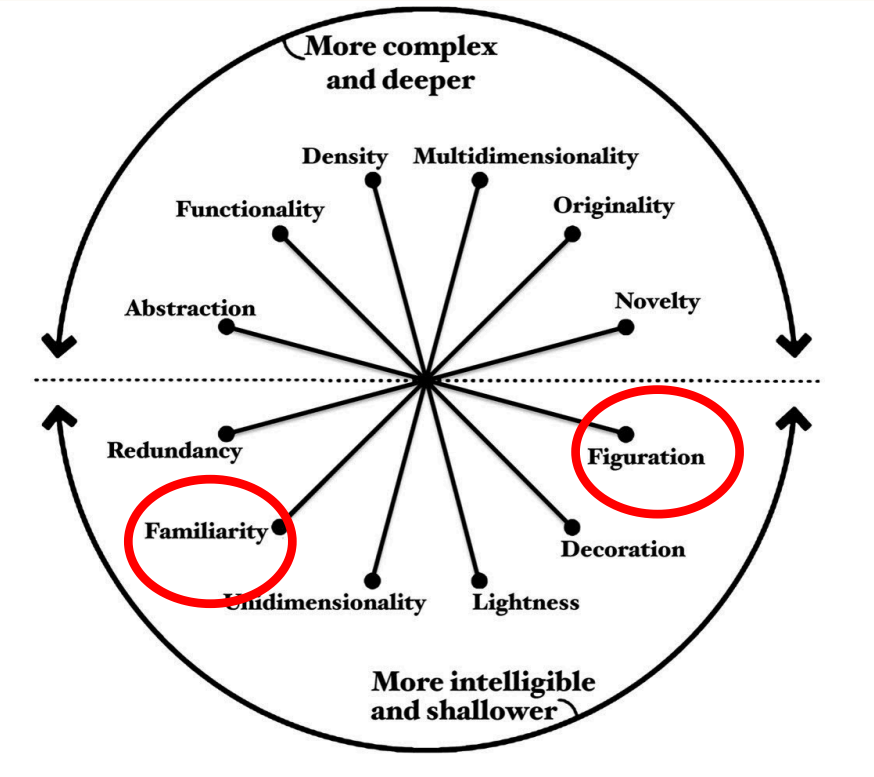
This map shows unemployment rates by U.S. county, August 2016. Darker blue means a higher rate.

Choropleths make **spatial distributions, clusters, and regional differences** easy to see. Compare the darker Southwest with the lighter central Plains.

Familiar boundaries help identify regions. Many statistical tables already report values by region.

Map: Mike Bostock / D3 · Data: U.S. Bureau of Labor Statistics · August 2016.

Choropleth maps: pros and cons



Familiarity: a known layout.
Figuration: recognizable shapes.

Visualization wheel: Alberto Cairo · The Functional Art.

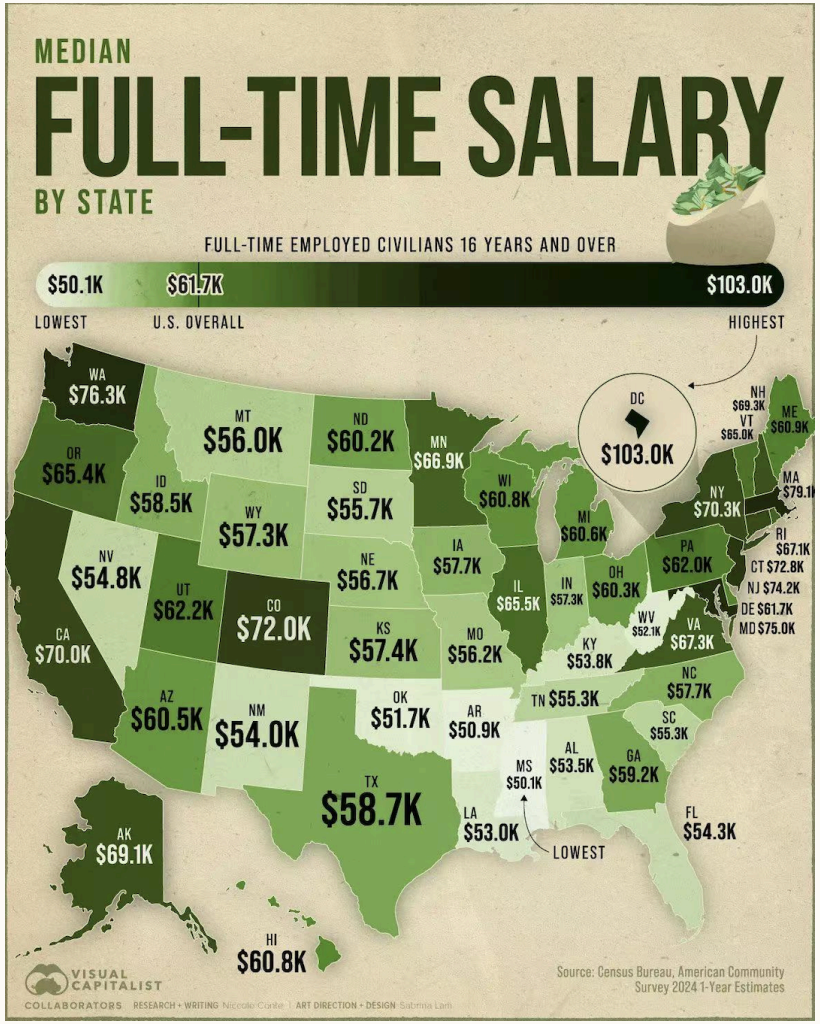
Pros

- Familiar map layouts and recognizable geographic shapes make regions easy to identify.
- Many datasets use geographic units, such as census areas.

Cons

- Position is fixed by geography, so it cannot encode another quantitative attribute.
- Large regions can dominate attention, hiding patterns in smaller regions.
- Colors and class intervals can change visible patterns.

Size can overwhelm color



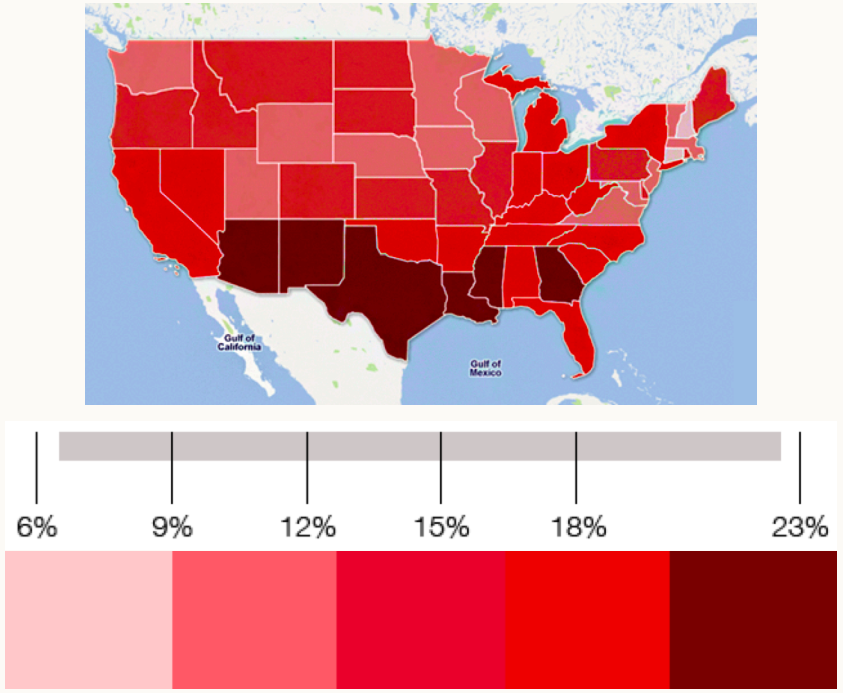
Separability means reading one channel without substantial interference from another.

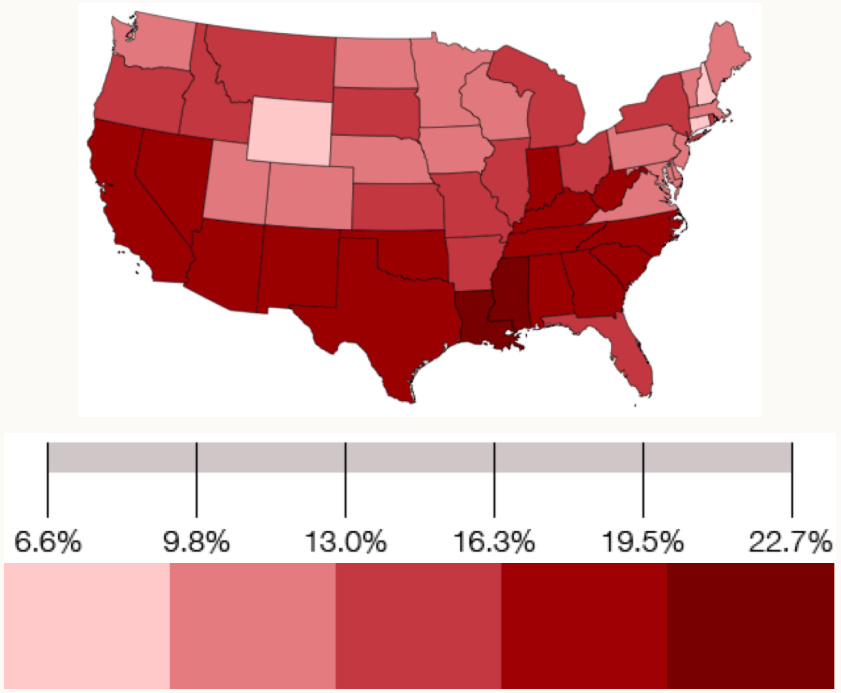
Here, **size can overwhelm color**. Large geographic regions attract attention, making the colors of smaller regions easier to miss.

Texas ($58.7K) occupies much more map area than Massachusetts ($79.1K) and Delaware ($61.7K), even though both have higher median salaries. **Size interferes with reading the salary encoded by color.**

Graphic: Niccolo Conte and Sabrina Lam / Visual Capitalist · Data: U.S. Census Bureau, ACS 2024 1-year estimates.

Class intervals strongly affect visible patterns

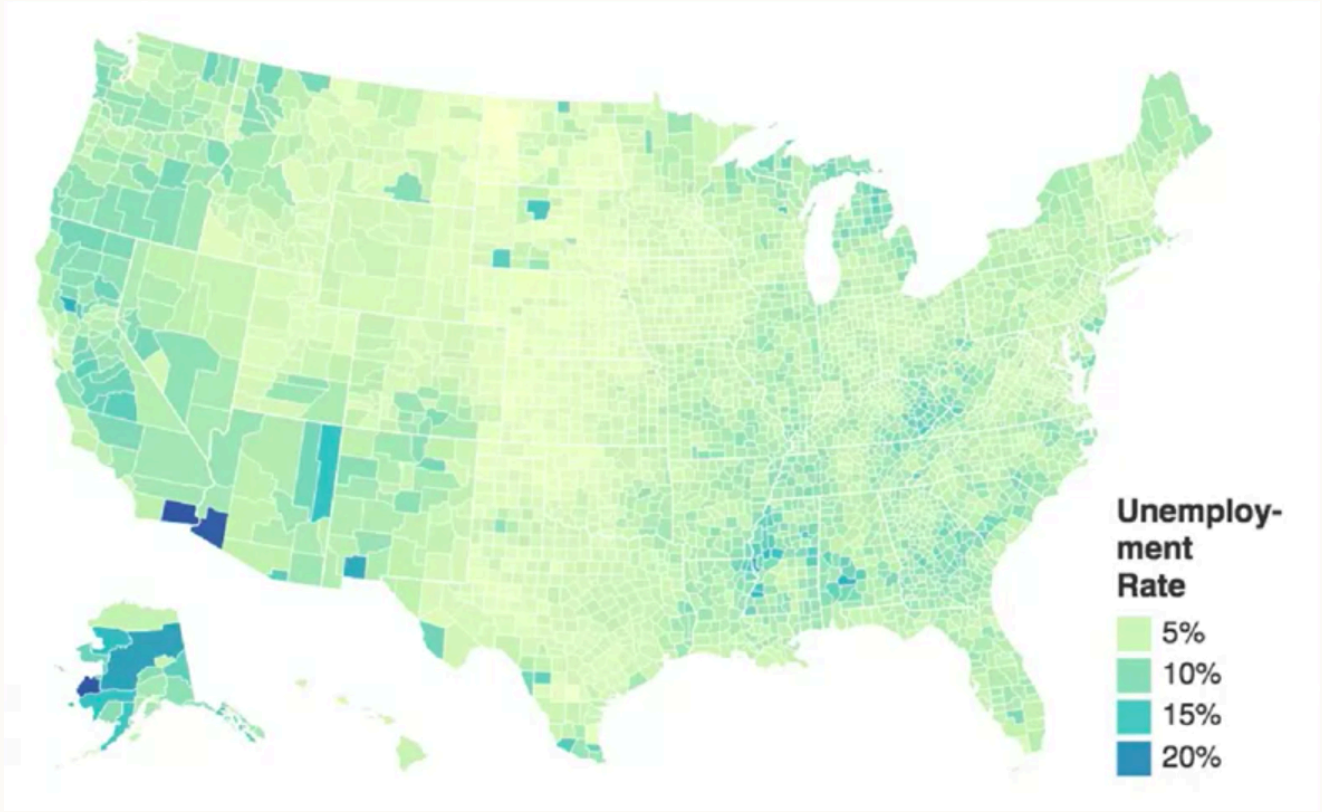




These maps visualize the same U.S. poverty dataset. **Left: unequal bin widths.** The first four bins span 3 percentage points each (6–9%, 9–12%, 12–15%, 15–18%), but the last spans 5 (18–23%). **Right: equal bin widths** of about 3.2 percentage points. These choices change which values share a color, producing different visible patterns.

Maps: Simon Rogers / The Guardian; comparison: Gregor Aisch (2011) · U.S. Census Bureau poverty rates (%).

A linear color scale can hide variation in skewed data



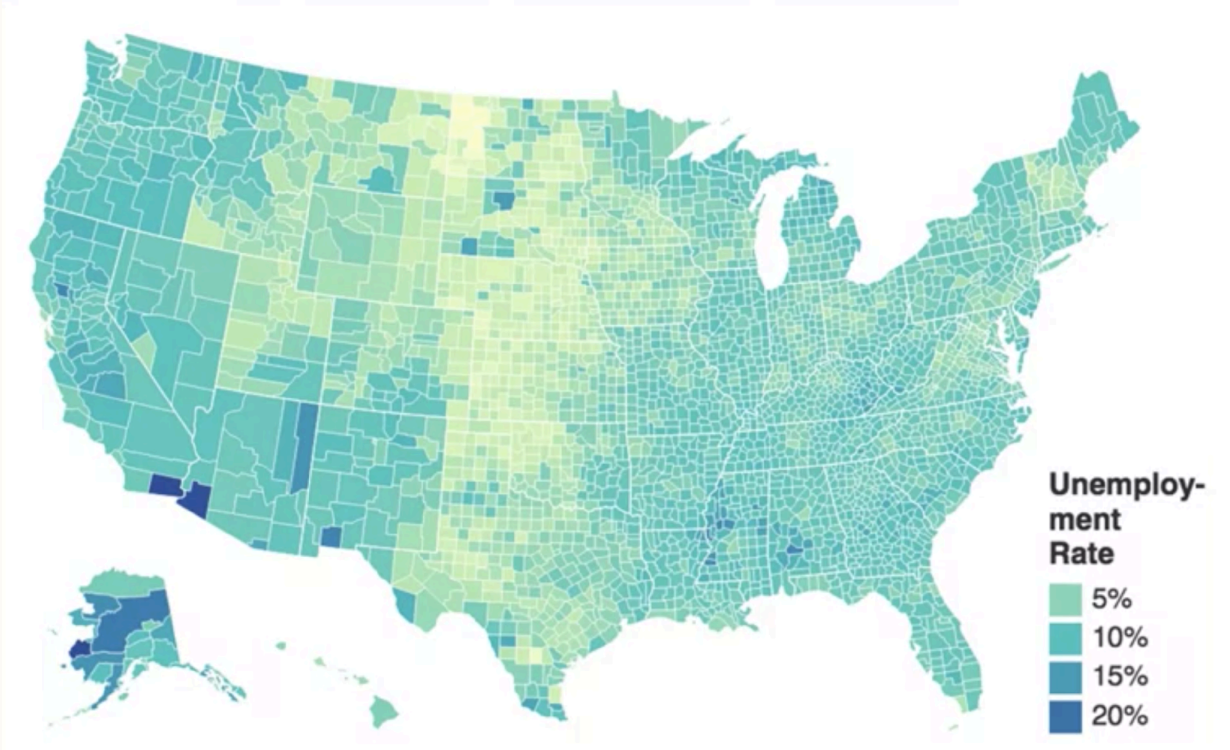
This map visualizes **U.S. county unemployment rates**. The distribution is **right-skewed**: many counties have low rates; a few have high rates.

A **linear color scale** spreads the colors across 1–26%. Low values receive similar pale colors, making their differences hard to see.

Equal-width class intervals can have a similar problem: many observations fall into a few bins, hiding variation within those bins.

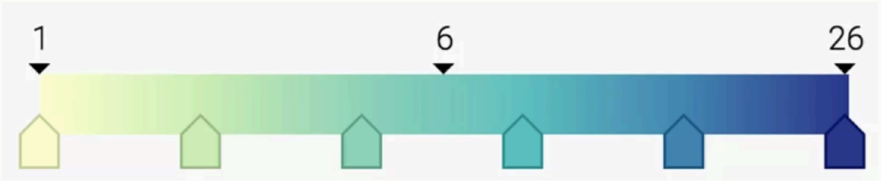
Map and color scale: Lisa Charlotte Muth / Datawrapper (2017) · Historical U.S. county unemployment example.

Choose intervals from the data distribution



The same data now places the **median, 6%, at the middle color**. About half the counties use each half of the color scale.

The narrower **1–6% range** receives more color variation, making differences among low-rate counties easier to see.



For a classed map, try quantile classification: choose class breaks so each interval contains roughly the same number of observations. Use narrower intervals where values cluster and wider intervals where they are sparse, so readers can see more detail in the crowded range.

Map and scale: Lisa Charlotte Muth / Datawrapper (2017) · Quantile classification explained (2022).

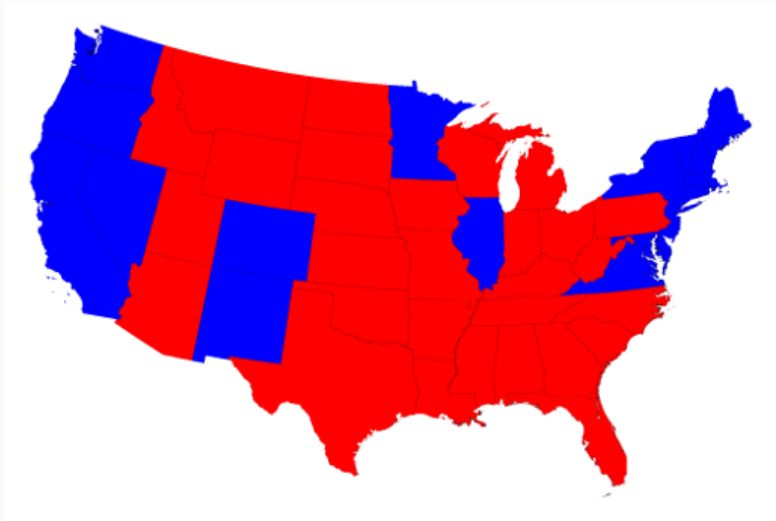
Variations of Choropleth map

- Cartograms
- Symbol map

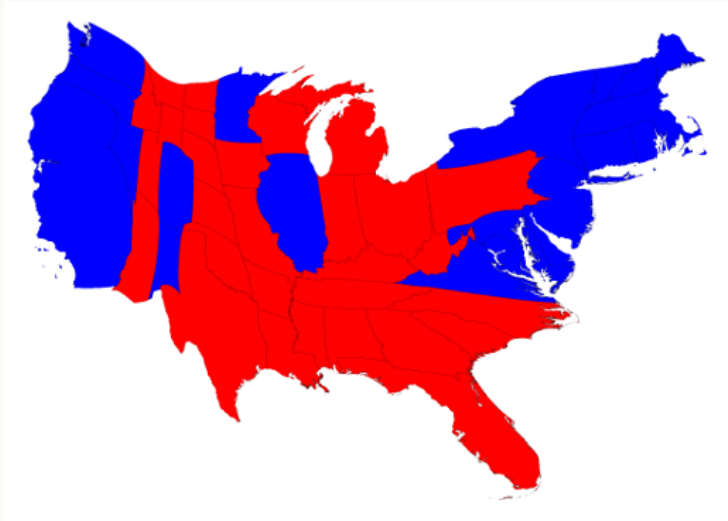
Cartograms resize regions to represent a quantity

An **area cartogram** resizes map regions so their **areas are proportional to a data attribute**.

Ordinary map: geographic area



Cartogram: population

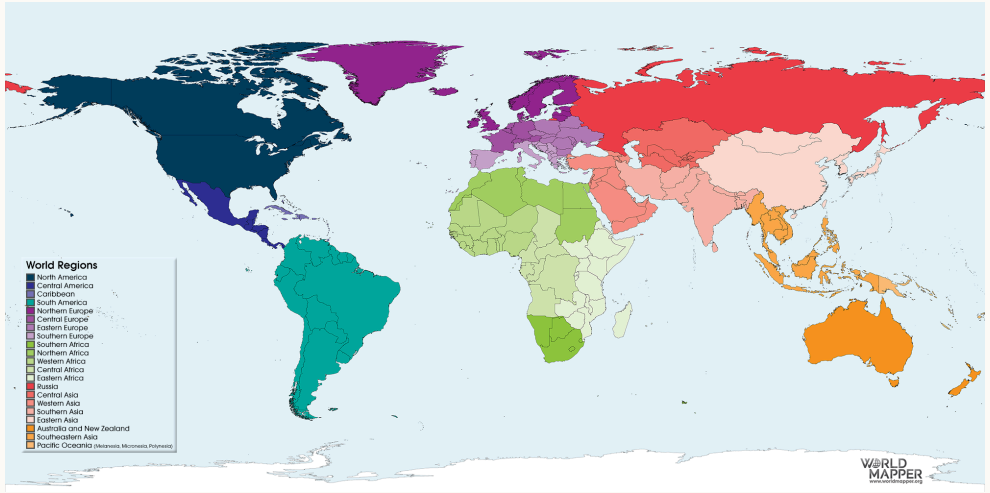


Both maps show the **2016 U.S. presidential election results**. **Left:** red covers most of the map, suggesting that most voters supported Republicans. **Right:** after resizing states by population, red and blue occupy similar areas; red no longer visually dominates.

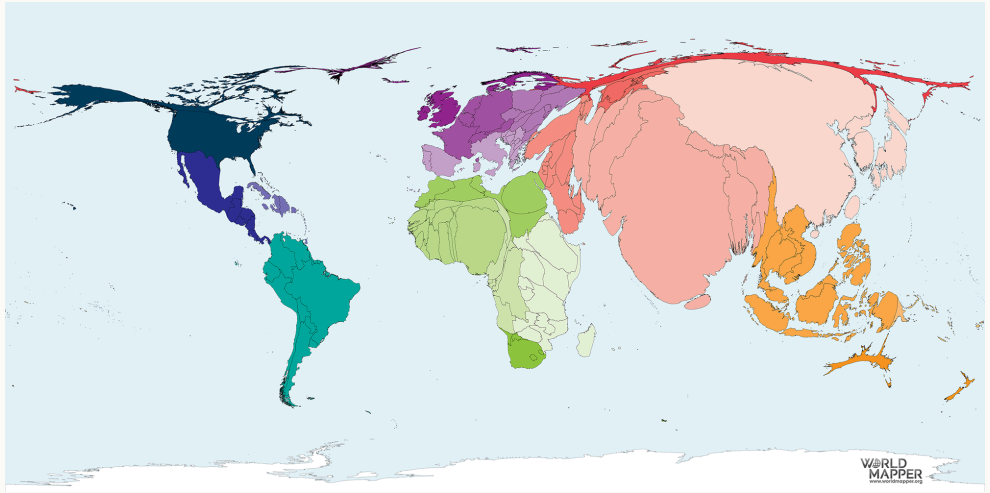
Maps: Mark Newman / University of Michigan · 2016 · Color = state winner: Republican (red), Democratic (blue), not vote share.

Cartograms reduce area bias but distort shapes

Ordinary world map



Population cartogram · 2018



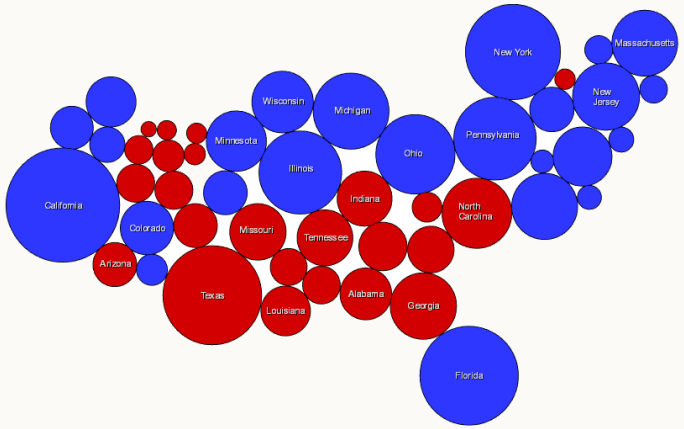
- **Advantage:** area represents the chosen attribute, so large geographic regions no longer dominate simply because of their land size.
- **Limitation:** distorted **area marks** can lose their familiar shapes, making regions difficult to recognize.
- **Mitigation: labels** identify regions when their shapes are no longer recognizable.

Maps: Worldmapper · Reference map / Population 2018 · Data: UN WPP, 2017 revision · CC BY-NC-SA 4.0.

Dorling uses circles; Demers uses squares

2012 U.S. election: each mark represents a state; its **area is proportional to total votes**.

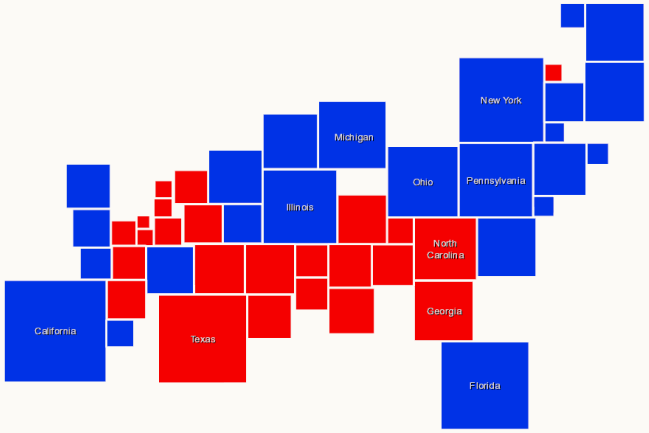
Dorling cartogram



Benefit: separate circles let us compare California and Texas without irregular outlines.

Cost: curved edges leave gaps. State shapes and exact positions are lost.

Demers cartogram



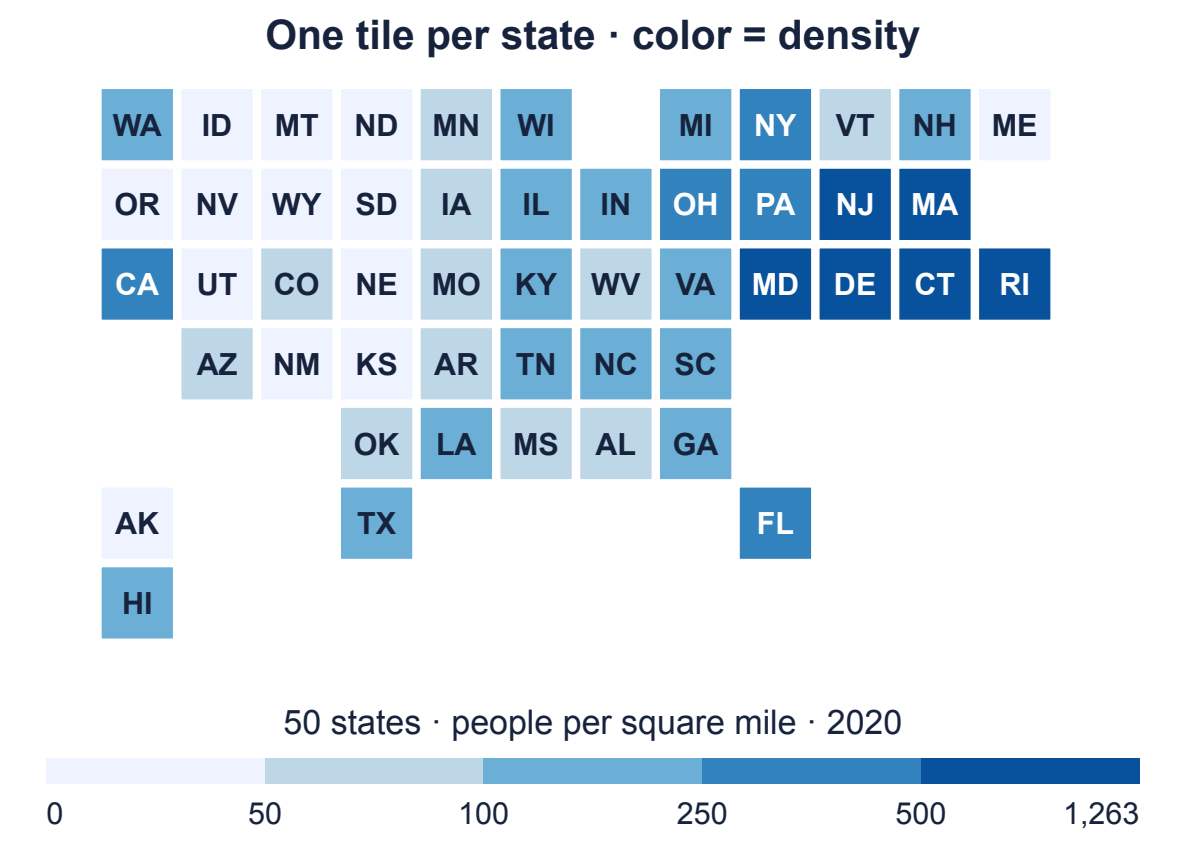
Benefit: squares pack more tightly, leaving smaller gaps between marks.

Cost: tighter packing can require larger shifts; contacts may not be real neighbors.

Maps: Kenneth Field · UCGIS, Figs. 3–4 · Blue = Democratic winner; red = Republican winner · Labels identify states.

Grid maps give every region equal space

An **equal-unit grid map** represents each region with an **equal-size tile**, arranged to approximate geography.



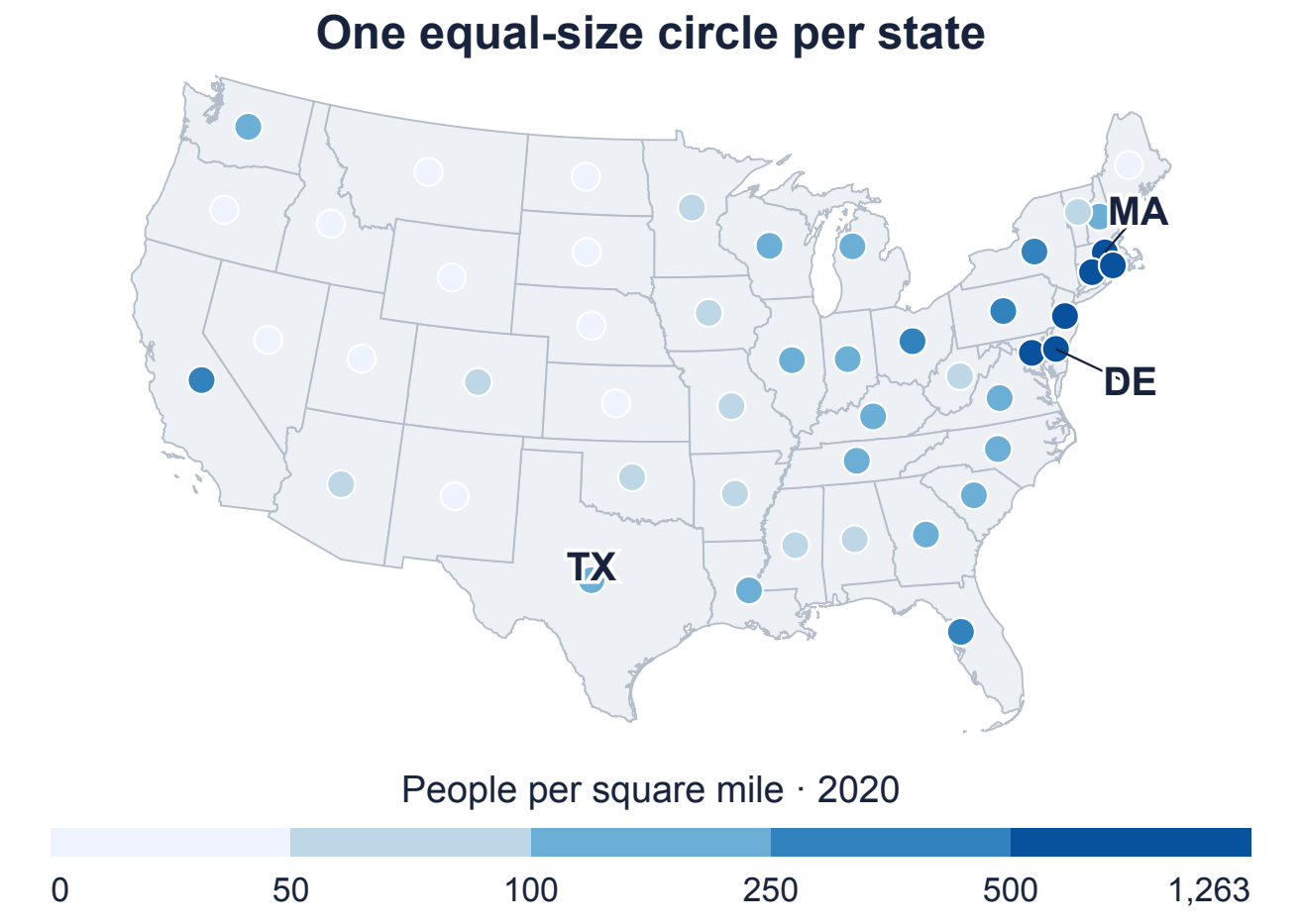
Data: U.S. Census Bureau · 2020 · 50 states · Course-designed approximate grid.

Example: one tile = one U.S. state; color = population density.

Advantage: every region gets equal display space, so small regions are easier to see and label.

Limitation: true shapes and exact locations are lost; tile neighbors may not be geographic neighbors.

Symbol maps separate color from geographic area



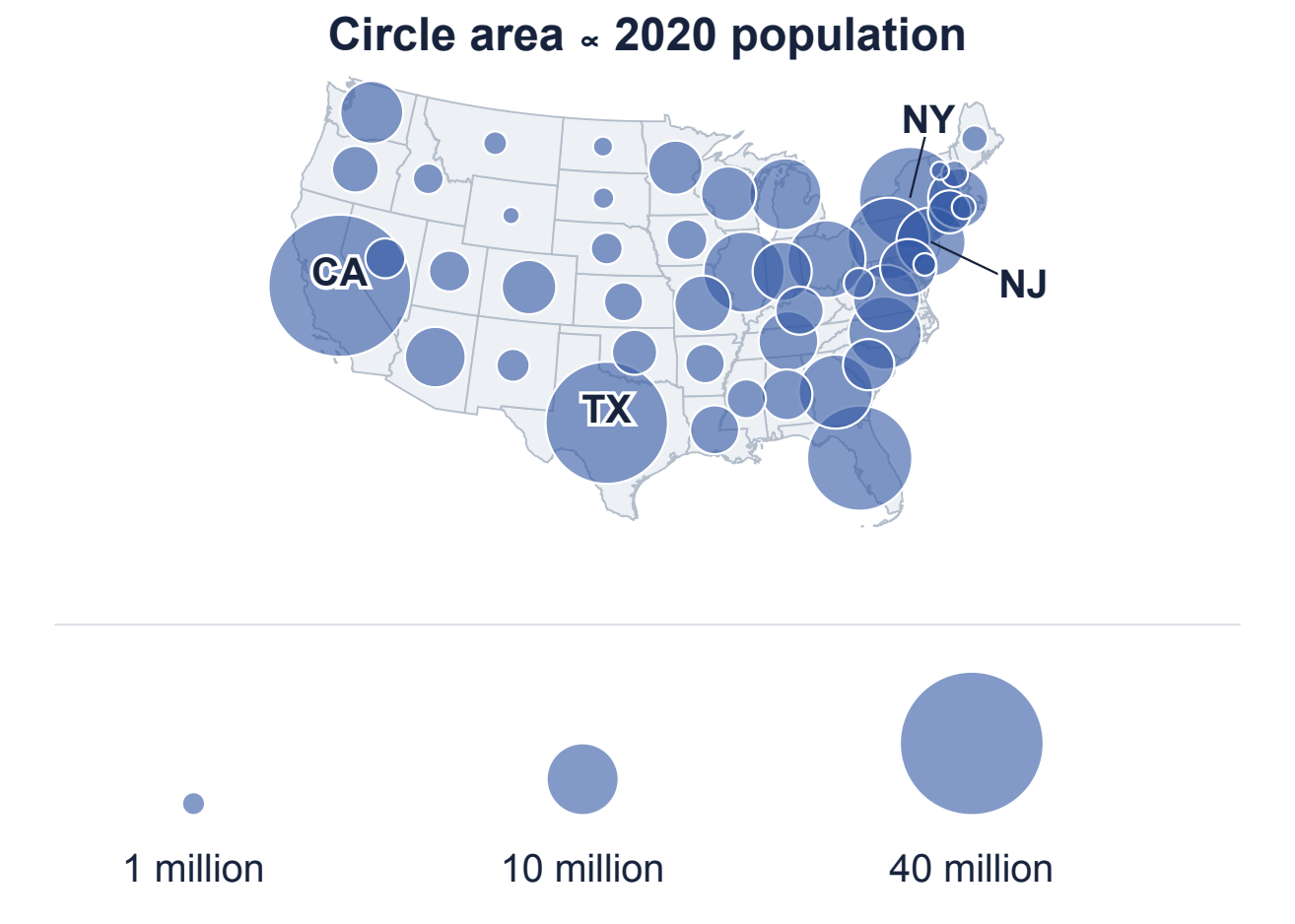
A **symbol map** places a symbol at a location or regional anchor and uses color, size, or shape to encode attributes.

Returning to the population-density example, every state gets an **equal-size circle**, with color encoding density.

Texas no longer receives more colored area simply because it has more land. The background still provides geographic context.

Data: U.S. Census Bureau · 2020 · Boundaries: us-atlas / Census 2017 · Contiguous 48 states.

Proportional symbols can overlap



A **proportional-symbol map** makes symbol area proportional to a quantity. Here, circle area represents 2020 population.

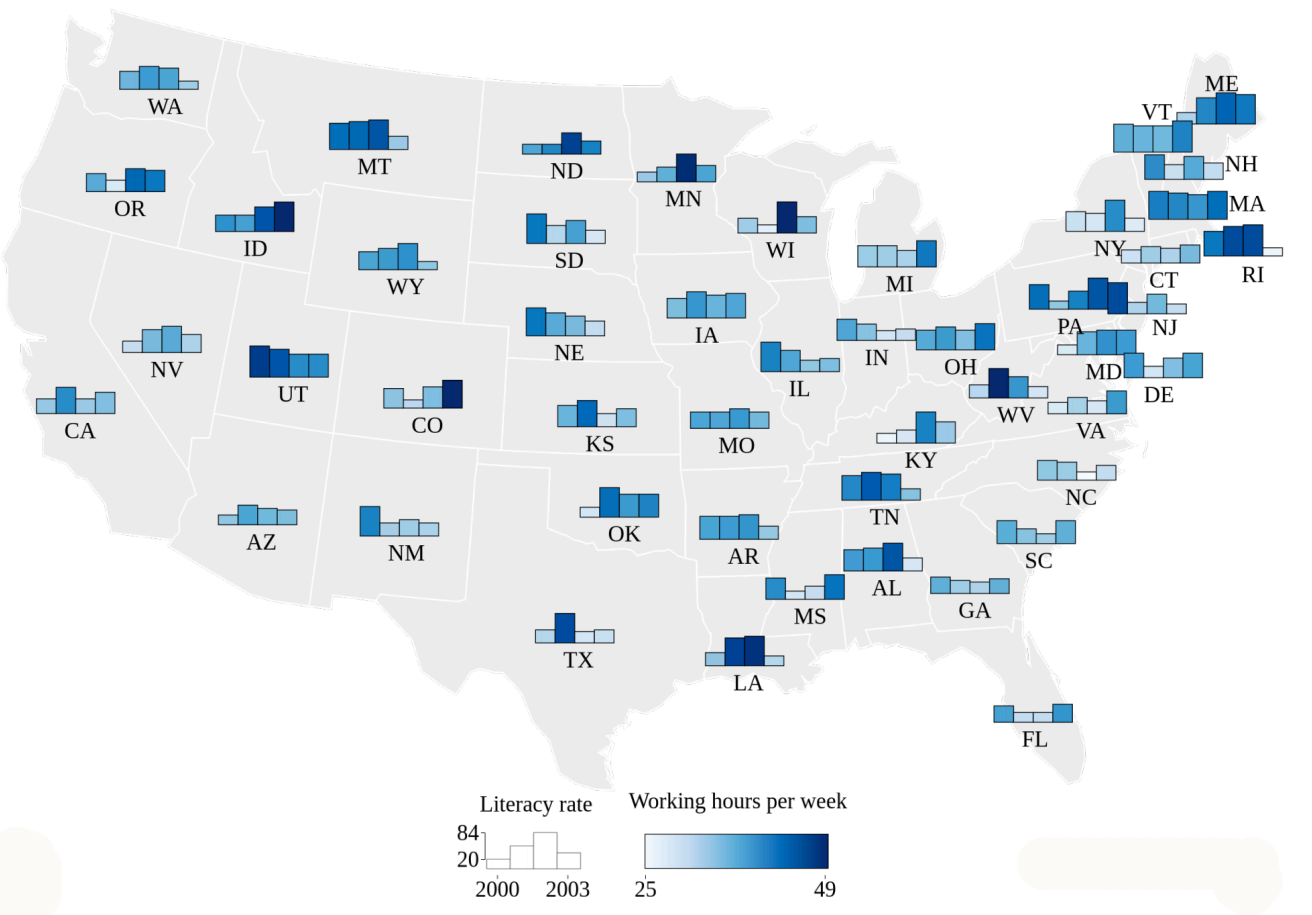
Around **New York and New Jersey**, nearby circles overlap and hide parts of one another.

Smaller symbols, transparency, zoom, or insets can help. Moving symbols apart improves visibility but changes their geographic positions.

Data: U.S. Census Bureau · 2020 · Boundaries: us-atlas / Census 2017 · Contiguous 48 states.

Glyphs encode multiple attributes at each location

A glyph combines marks or channels to encode **multiple attributes of one data item**.



Source: Peña-Araya, Pietriga & Bezerianos · TVCG (2020), Fig. 1c · Artificially generated data.

Each state uses four bars for 2000–2003, ordered left to right.

Bar height shows literacy rate; darker blue shows longer working hours per week.

In Texas, the second bar is taller and darker: both variables are higher in 2001.

Glyphs show local changes over time, but can overlap in crowded regions.

PART 03

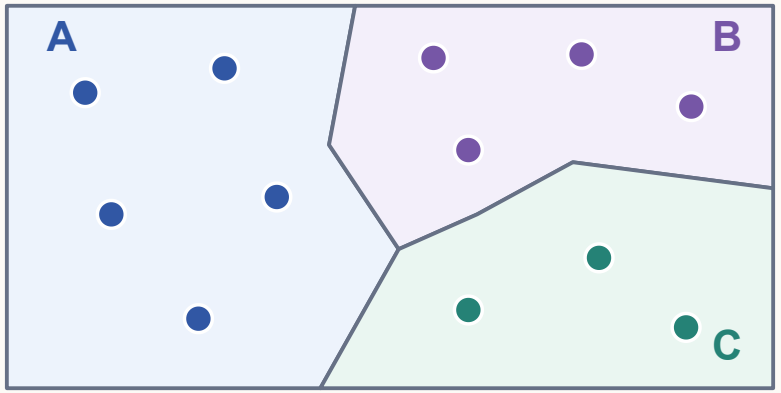
MAUP: aggregation boundaries change the result

The same observations can produce different statistics and map patterns when we change how space is divided.

Spatial aggregation summarizes data within regions

Spatial aggregation groups observations by the regions they fall within, then computes a summary—such as a count, total, mean, or rate—for each region.

Before: 12 individual incidents



Each dot is one incident at a recorded location.

After: 3 regional summaries

Each region has one value: its incident count.

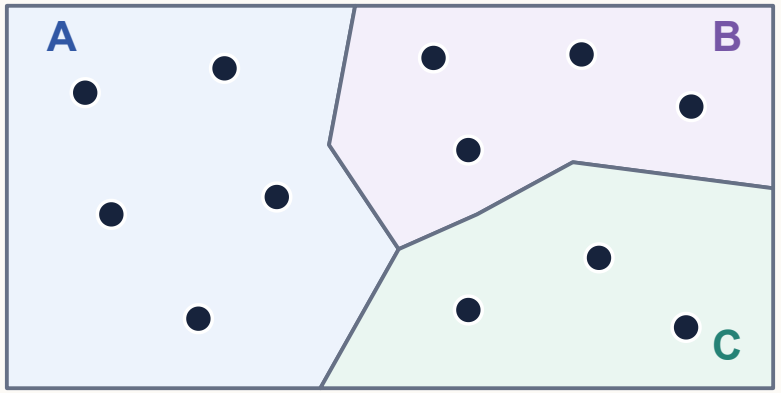
Assign each point to a region, then count: A = 5, B = 4, C = 3. All 12 incidents are included, but their individual locations are no longer shown in the aggregated result.

Hypothetical teaching example · Regions A–C are schematic; colors identify regions, not incident counts.

MAUP: boundaries change aggregation results

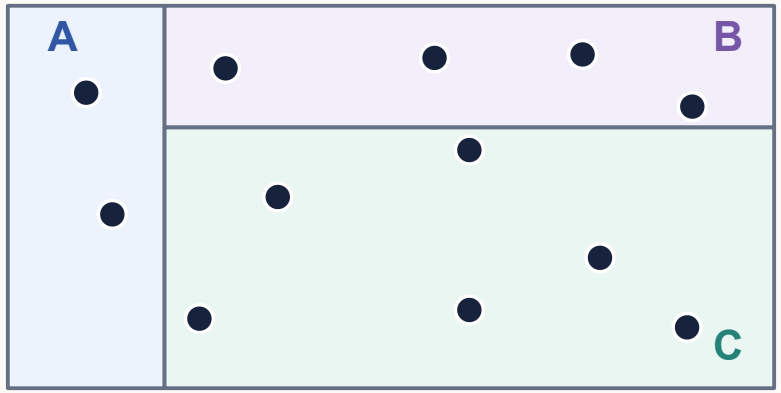
The **modifiable areal unit problem (MAUP)** occurs when statistical results change because we change the size or boundaries of the areas used to aggregate the **same underlying observations**.

Original regions



Counts: **A = 5** · B = 4 · C = 3

Redrawn regions



Counts: A = 2 · B = 4 · **C = 6**

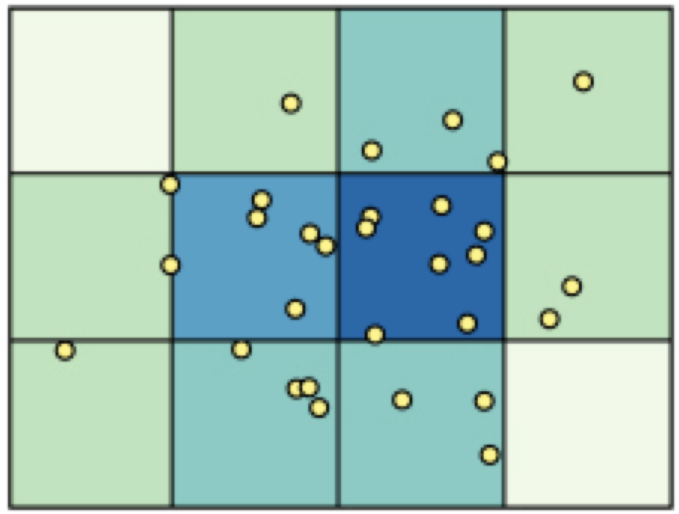
MAUP has two effects: **zone effect** (redrawing boundaries) and **scale effect** (changing the size or number of regions). This example shows the **zone effect**: the same 12 points produce different regional counts.

Hypothetical example · Same 12 incidents as the previous slide · Definition: Esri GIS Dictionary.

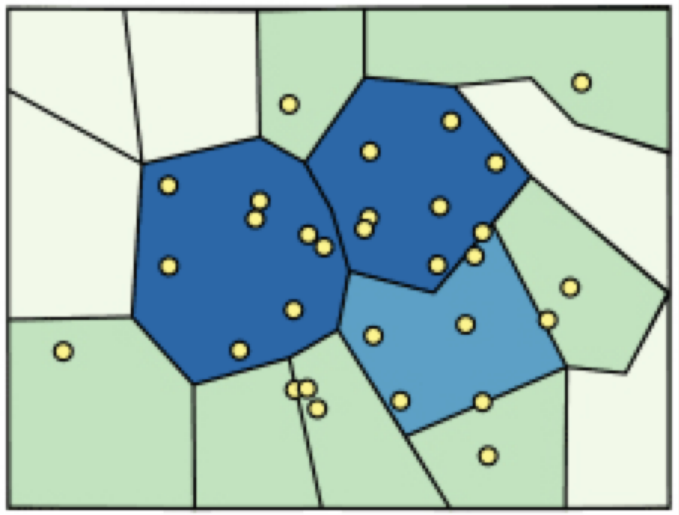
Zone effect: redraw boundaries, change the pattern

The **same point locations** are aggregated into three different sets of zones. Darker blue represents higher point density within a zone.

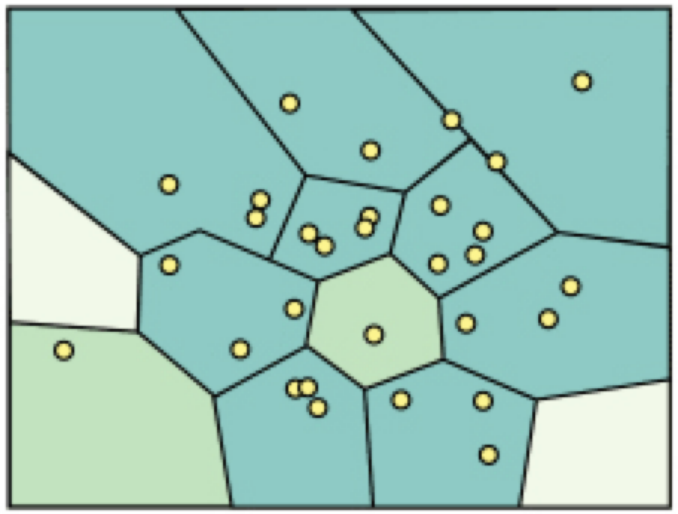
Regular grid



Cluster grouped



Cluster split



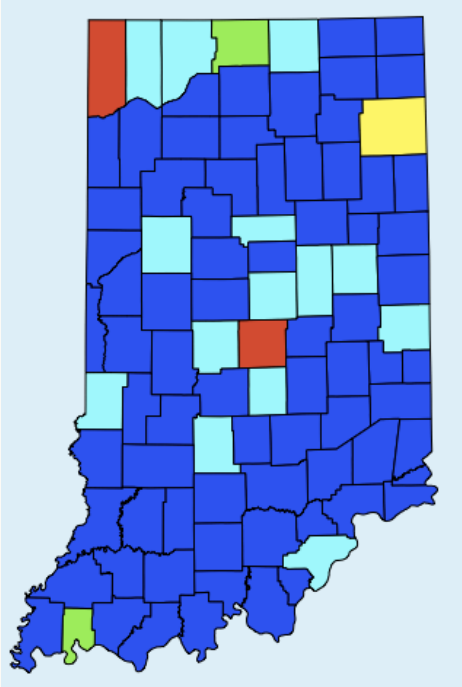
Grouping nearby points together makes the central concentration stand out. Splitting those points among zones with less-dense surroundings makes the pattern look more uniform.

Illustrations: Penn State GEOG 486, Fig. 4.cg.6 · Attribution reproduced in Tamara Munzner's visualization course.

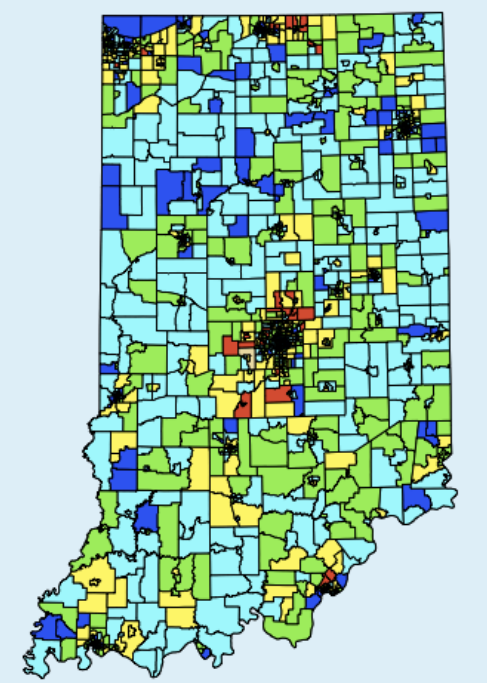
Scale effect: counties combine census-tract detail

The **scale effect** arises when the same observations are summarized using larger or smaller geographic units.

Counties



Census tracts



Households in Indiana

Each county total combines households from many smaller census tracts.

A county receives one color, so **differences among its tracts disappear** from the county map.

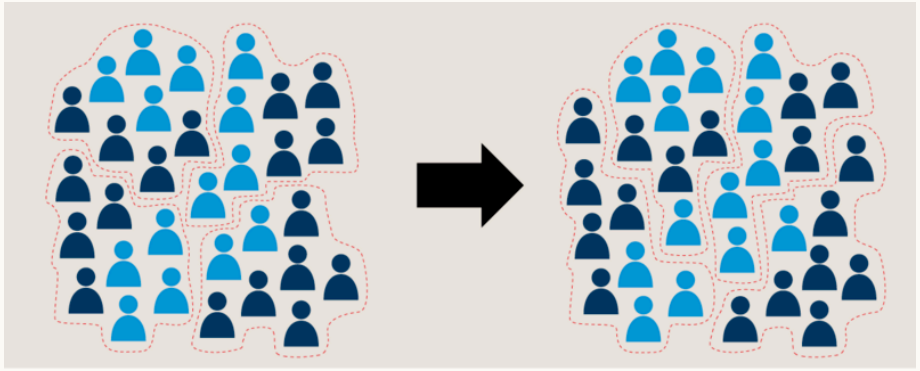
The tract map retains finer detail. Matching colors across these two maps should **not be read as equal counts**.

Maps: Rick Jones / Cartographica (2011) · Data: IndianaMap · The Modifiable Areal Unit Problem in GIS.

Packing and cracking are gerrymandering techniques

Gerrymandering is the intentional manipulation of electoral district boundaries to give a political group an electoral advantage. In other words, the voters stay in the same places, but changing their district membership changes the aggregated election results.

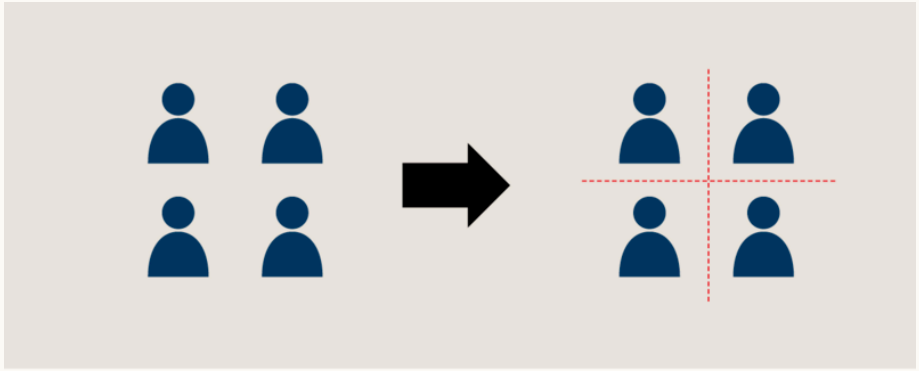
Packing



"PACKING"

Concentrate opponents in a few districts. They win by large margins there, leaving fewer supporters to win elsewhere.

Cracking

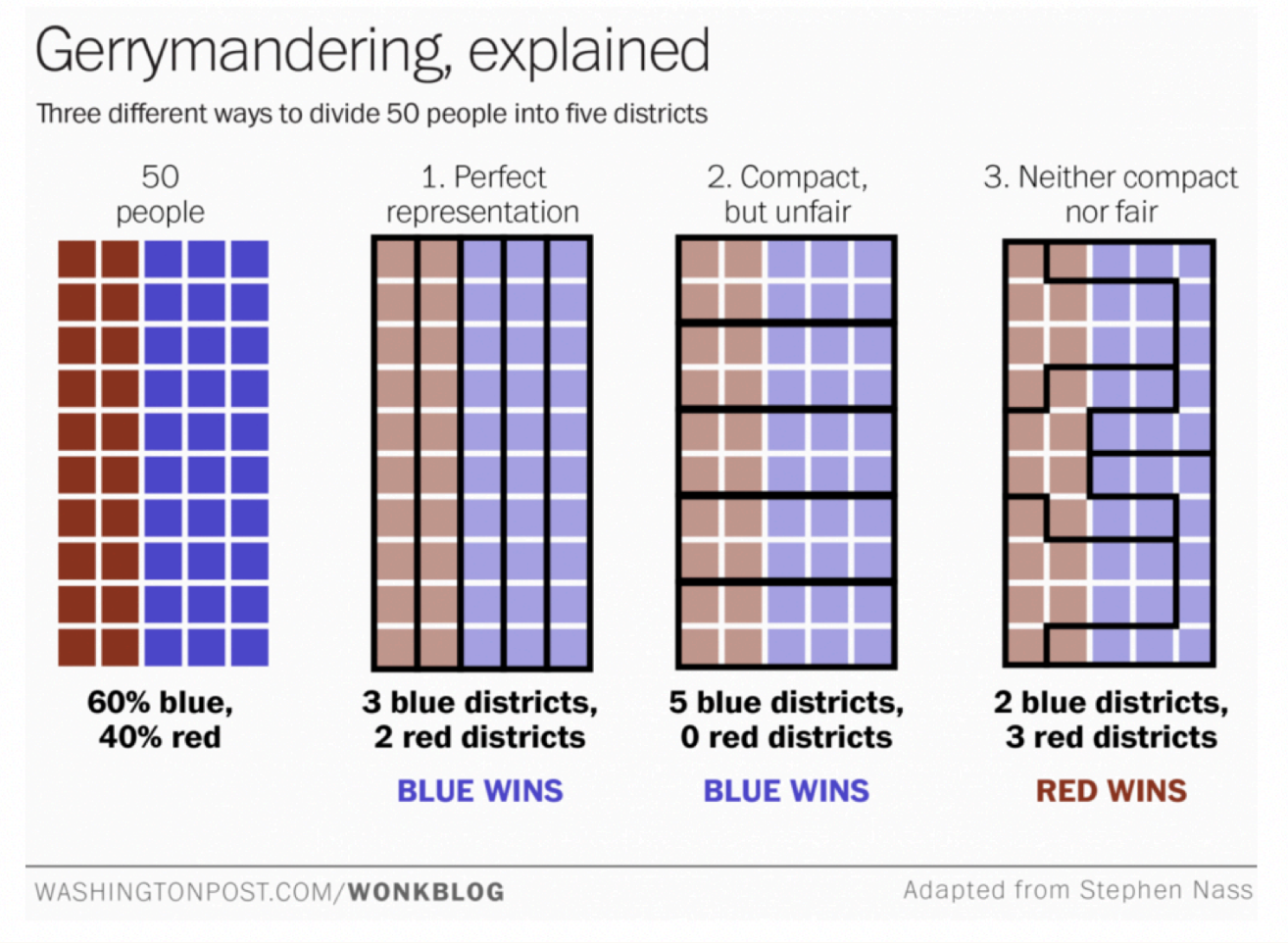


"CRACKING"

Split opponents across several districts. They become minorities in enough districts to reduce the seats they can win.

Definitions and diagrams: Brennan Center for Justice · Gerrymandering Explained · Schematic voters.

The same voters can produce different seat totals



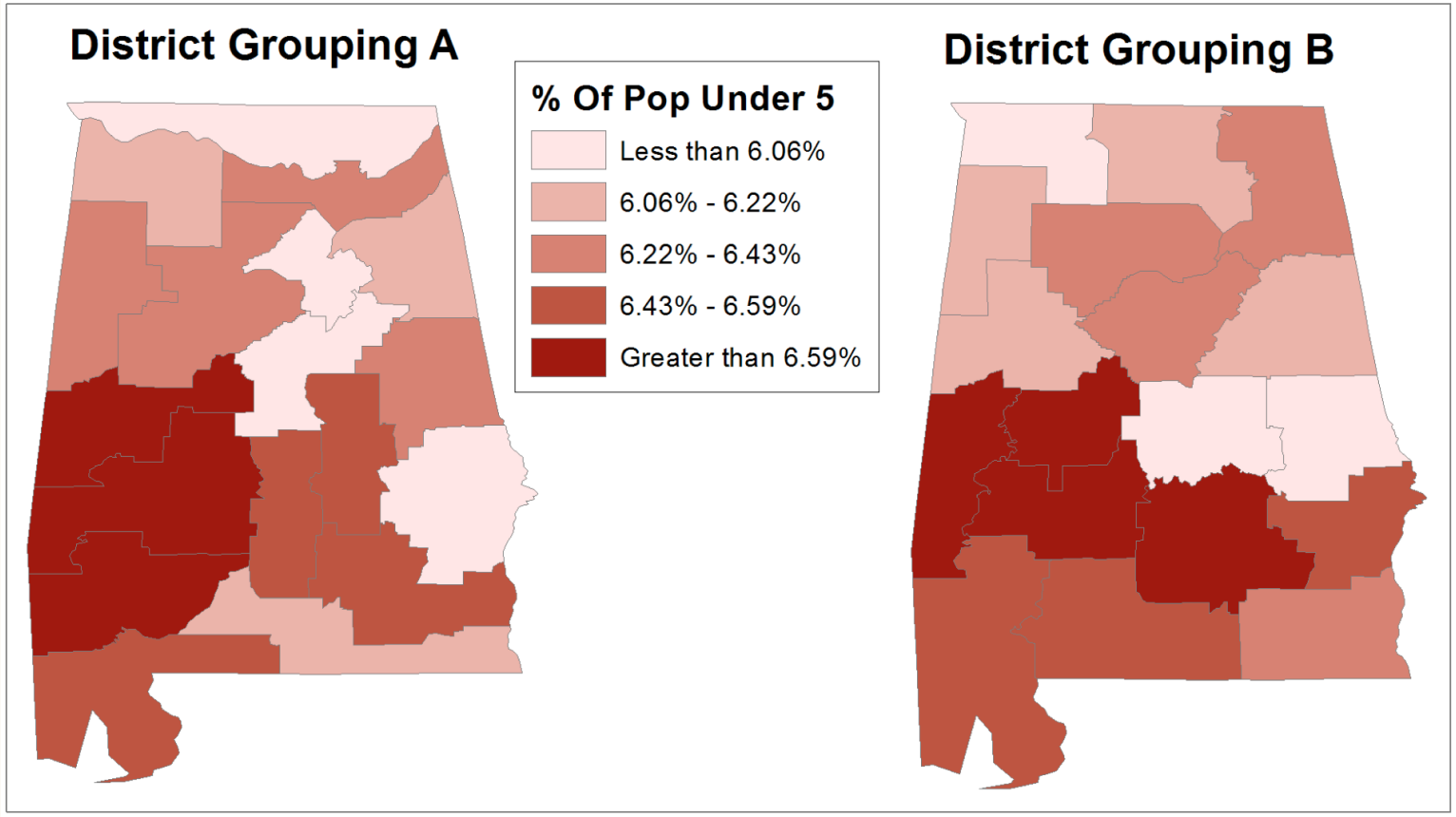
In this hypothetical example, **30 voters support blue** and **20 support red**. Each district contains 10 voters and elects one representative.

Changing only the boundaries gives blue **3, 5, or 2 of the 5 seats**.

In the last plan, many blue voters are **packed into two districts**; red wins a majority in the other three.

Illustration: Christopher Ingraham / The Washington Post (2015), adapted from Stephen Nass · Hypothetical voters.

A regional summary can hide local differences



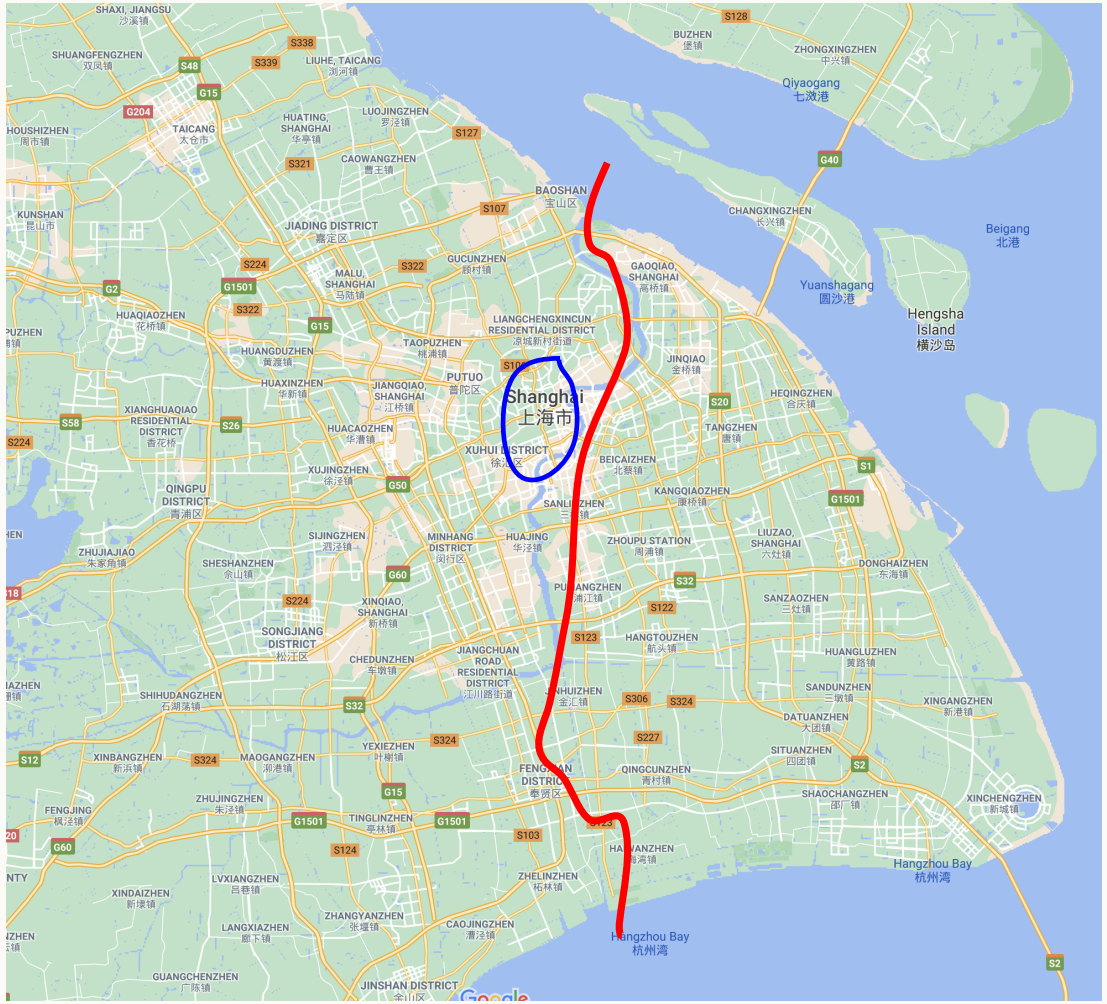
These maps show the **percentage of residents under age 5 in Alabama**.

The same counties are combined into two different sets of larger regions, using the **same color classes**.

A county's regional color changes with its **aggregation partners**. One regional value also hides differences among its counties.

Map: Efram Stone / METRANS (2017), Fig. 3 · Common legend in both maps.

Choose aggregation units that fit the question



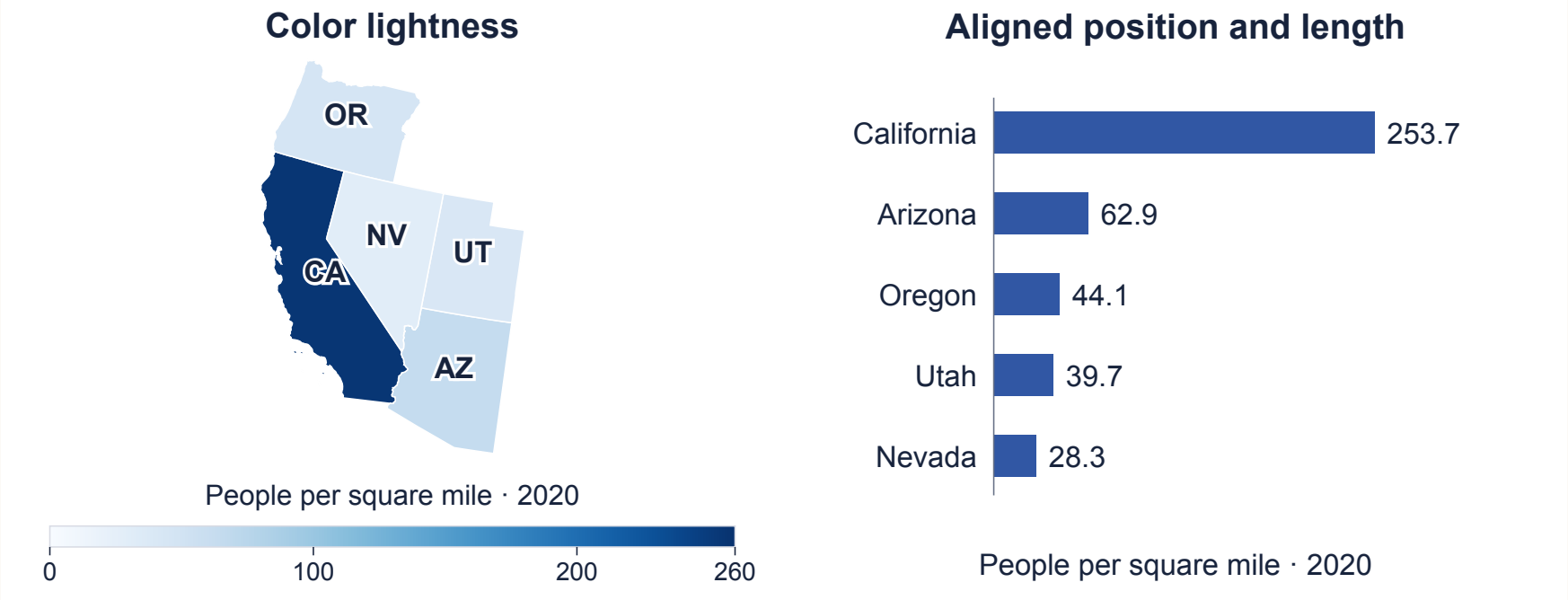
Example task in Shanghai	Possible units	Why these units?
Compare district-level service budgets	Administrative districts	Match the level responsible for the budget.
Locate gaps in neighborhood services	Subdistricts or smaller areas	Retain local variation hidden by district totals.
Compare spatial event densities	Equal-area grid cells	Use a consistent geographic footprint.

Zones should be **easy to understand** and **rational** to audience and tasks.

Map: Google Maps · Colored lines are schematic, not exact boundaries. Guidance: UCGIS · Boundaries and Zone Membership.

Other considerations for map layouts

Color ranks below position and length

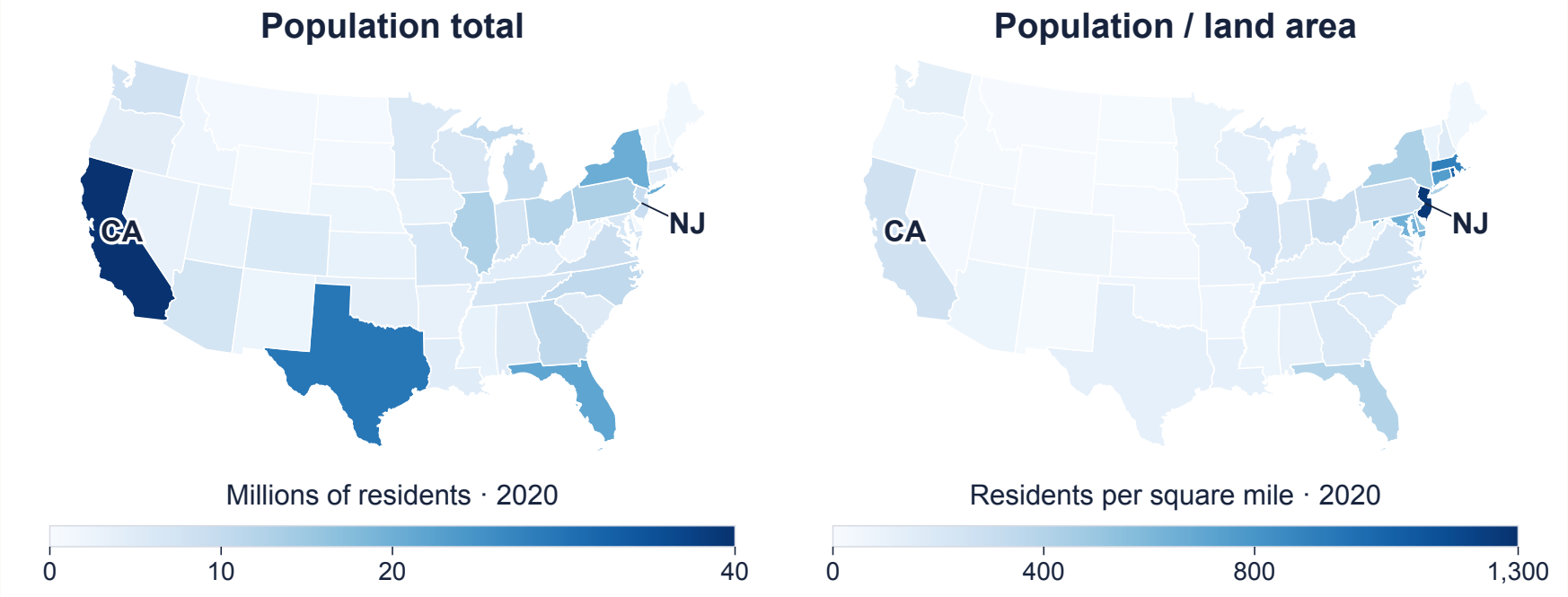


Channel ranking: for quantitative comparison, color lightness ranks below position on a common scale and length. Oregon (44.1) and Utah (39.7) have similar shades, so their small difference is hard to judge. **Aligned bar endpoints and lengths** make the difference easier to see.

Data: U.S. Census Bureau · 2020 · Boundaries: us-atlas / Census 2017 · Five western states.

Other considerations for map layouts

Totals and density answer different questions



California: 39.54 million residents, 253.7 people/mi². **New Jersey:** 9.29 million, 1,263 people/mi². Totals answer “how many”; dividing by land area answers “how concentrated.” Choose a denominator that matches the question.

Data: U.S. Census Bureau · 2020 · Boundaries: us-atlas / Census 2017 · Contiguous 48 states.

Maps: strengths and limitations

What they show

Spatial patterns. Compare regional clusters, gradients, and neighboring areas when geography matters.

One value per region. A standard choropleth uses fill color to encode one quantitative attribute. To show multiple values per region, use glyphs.

Totals or intensity. Use proportional symbols for totals, and choropleths for rates, proportions, or densities, according to the task.

What needs care

Color has limited precision. It ranks below aligned position and length for judging small numerical differences.

Color mapping changes patterns. The palette and class intervals affect which differences stand out. Check the data distribution.

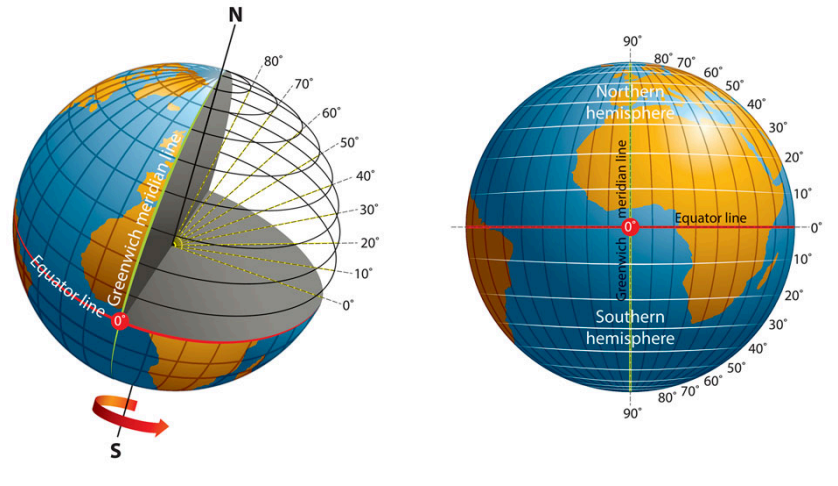
Layout trade-offs. Equal-size symbols reduce geographic-area bias but may overlap. Cartograms resize regions by data but distort their shapes.

MAUP. Boundaries and aggregation scale can change patterns and conclusions. Compare results across alternative groupings.

Appendix

Coordinates, projection families, and their trade-offs.

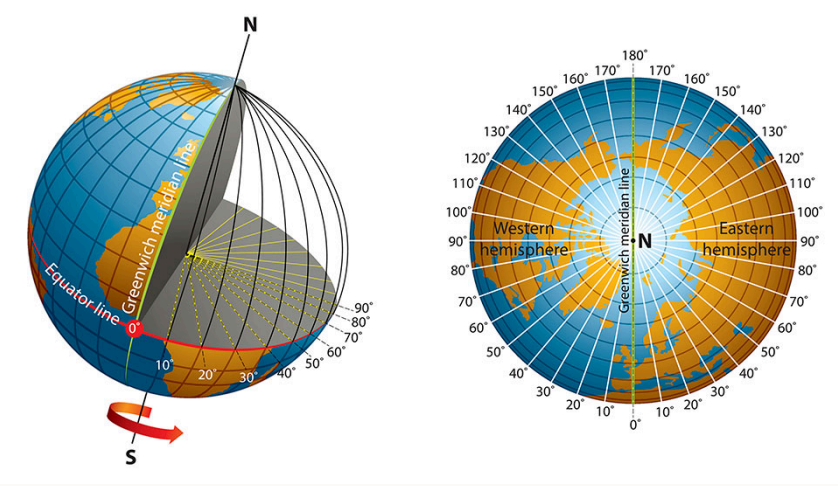
Latitude and longitude locate a point on Earth



Latitude: north or south

Latitude is an angle that locates a point north or south of the **Equator (0°)**. It ranges from **90°S to 90°N**.

Lines of equal latitude are called **parallels**.



Longitude: east or west

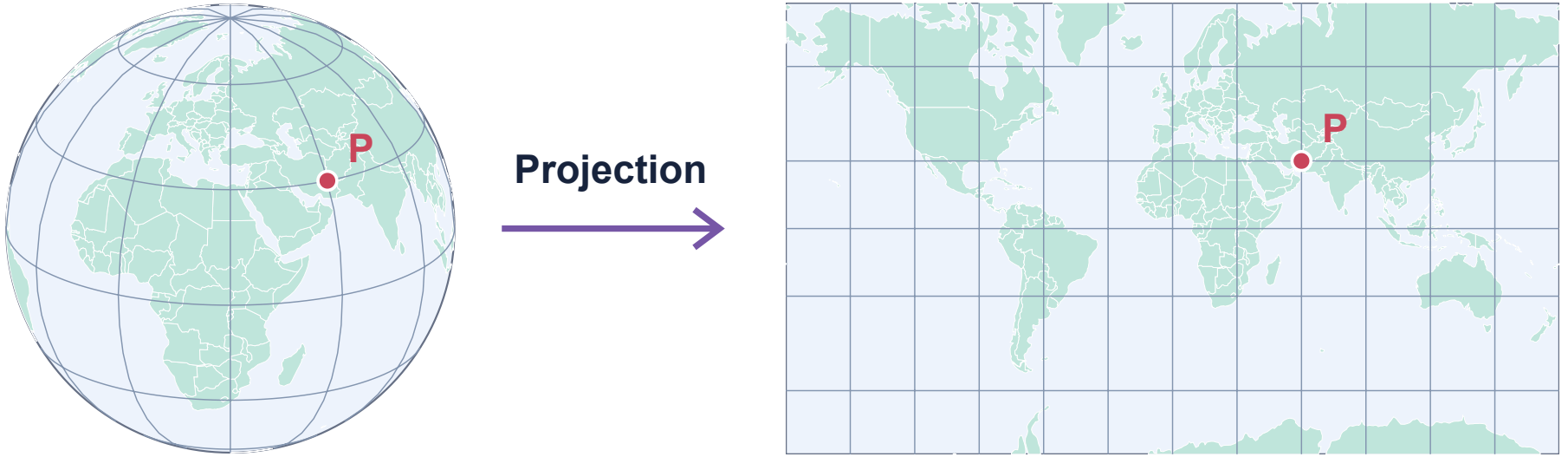
Longitude is an angle that locates a point east or west of the **prime meridian at Greenwich (0°)**. It ranges to **180°E or 180°W**.

Lines of equal longitude are called **meridians**.

Diagrams: NOAA / iStock · Latitude and longitude.

Why a flat map needs a projection

Earth is curved, while paper and screens are flat. A **map projection** is a mathematical rule that converts longitude and latitude into positions on a plane.



Globe: longitude and latitude

Flat map: x and y

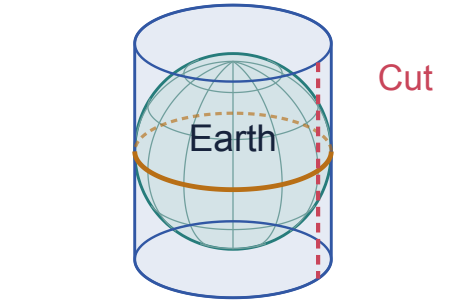
P stays at 60°E, 30°N. The projection determines where it is drawn.

Flattening changes distances, areas, or shapes. **No flat world map preserves all of them everywhere.**

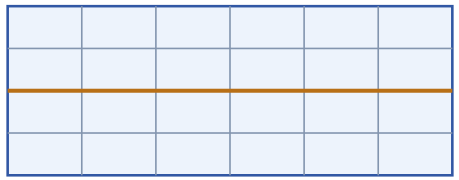
USGS · Map projections · Geometry: Natural Earth / World Atlas.

How projection surfaces become flat maps

Cylinder

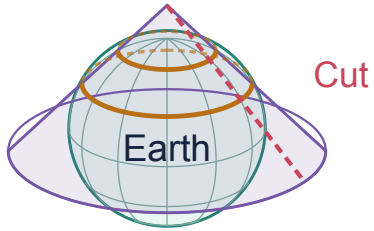


Touches the Equator
Cut and unroll

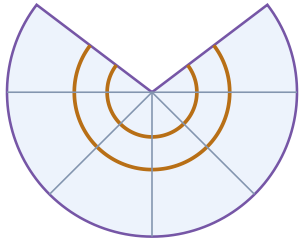


Rectangle

Cone

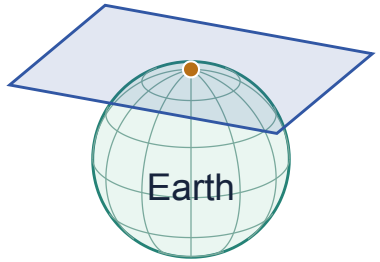


Cuts two parallels
Cut and open

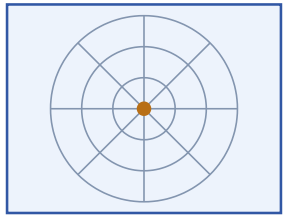


Circular sector

Plane



Touches one point
No unrolling needed



Already a flat surface

These surfaces are **geometric models**. Formulas determine how coordinates are placed on them.

Surface diagrams: course illustration · USGS · Map Projections.

Projections preserve different properties

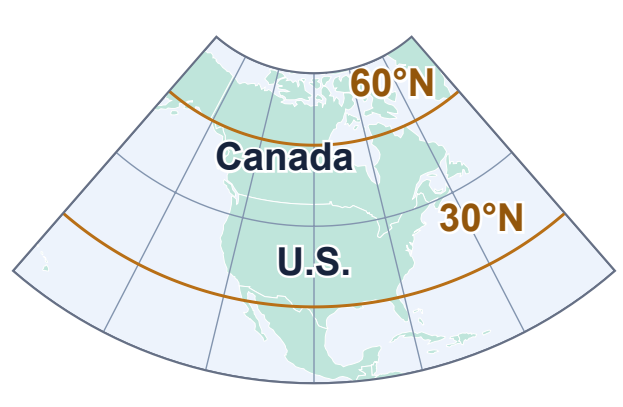
Mercator

Preserves: local angles and shapes.

Distorts: Greenland’s area is inflated relative to Africa.

Use: plot constant-bearing sea routes as straight lines.

Albers equal-area

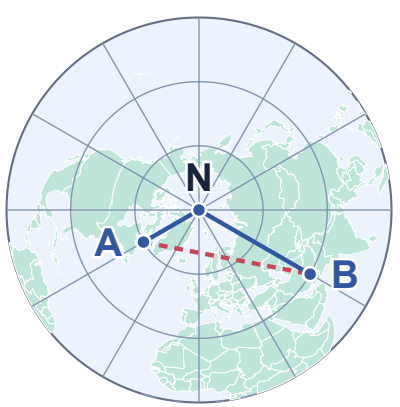


Preserves: relative land areas across North America.

Distorts: shapes and distances away from 30°N and 60°N.

Use: compare areas of land-cover regions.

Azimuthal equidistant



Preserves: distances and directions from N.

Distorts: shapes, areas and the A-B distance.

Use: compare distances from the North Pole.

Projection properties: Esri · Mercator · Albers · Azimuthal equidistant · Maps: D3 / Natural Earth / World Atlas.